

# Same Scene, Different Story: How Well Do Pretrained Perception Models Actually Work on Robots?

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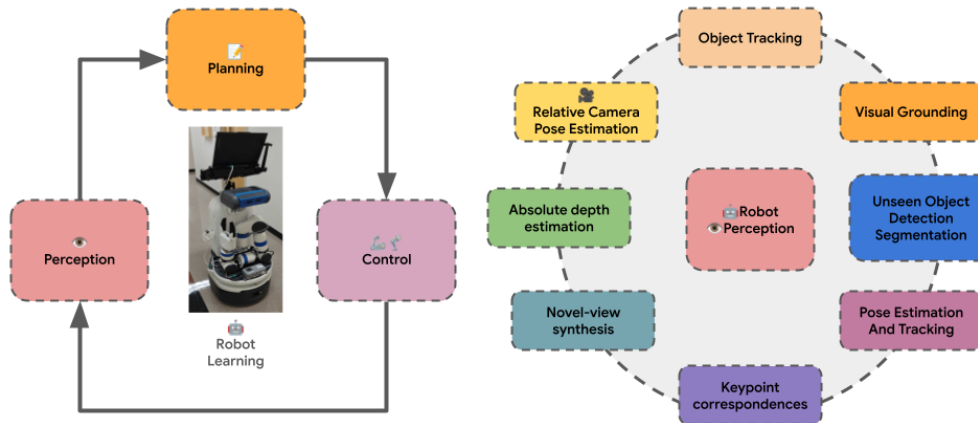


Figure 1: **RPX** framework overview. A human-carried rig (D435 RGB-D + T265 6-DoF pose + GoPro ego) captures 100 real-world scenes under a three-phase protocol (clutter → interaction → clean). Five perception tasks (§5) are evaluated on identical scenes and objects; the dataset additionally supports tasks shown in the figure but not evaluated in this work (§7). Deployment-readiness metrics (TS, STR, SGC) and Effort-Stratified Difficulty (ESD) reveal how foundation models perform under embodied deployment conditions—not just on static accuracy.

## Abstract

1       Pretrained perception models—depth estimators, object detectors, trackers, and  
2       vision-language models—are now standard components in robot learning systems,  
3       yet the choice of which model to deploy is made without evidence from the con-  
4       ditions robots actually face. Standard benchmarks use curated internet images;  
5       robots see the world through low-resolution Intel RealSense sensors under clut-  
6       ter, hand occlusion, motion blur, and challenging surfaces. We introduce **RPX**,  
7       a real-world evaluation methodology and benchmark that quantifies the *deploy-*  
8       *ment readiness* of pretrained perception models for embodied robotics. A human  
9       carrying the same sensor rig a mobile robot would use (D435 RGB-D + T265  
10      tracking + GoPro egocentric) captures 100 scenes spanning indoor and outdoor  
11      environments, daylight and night, reflective and transparent surfaces, across  $\sim 70$   
12      object categories and 75,000 frames. Each scene is captured in three phases that  
13      mirror the robot manipulation cycle: **clutter** (the scene as a robot naturally finds it),  
14      **interaction** (a human manipulates objects—the data that learning-from-humans  
15      pipelines train on), and **clean** (the scene reorganised—the condition models are  
16      designed for). The gap between clean-scene and clutter/interaction performance *is* the  
17      deployment readiness gap, and no existing benchmark measures it. **RPX** evaluates  
18      **TODO: N** models across **TODO: M** perception tasks on the *same* physical scenes  
19      and objects, using deployment-readiness metrics—State-Transition Robustness,

Geometric Coherence, and Effort-Stratified Difficulty—that capture properties invisible to static accuracy and compose into an Embodied Deployment Score (EDS). **TODO: HEADLINE FINDING.** **RPX** is released as a pip-installable, bring-your-own-model toolkit at <https://anonymous.4open.science/r/RPX>.

## 1 Introduction

What the robot fails to see, it fails to do. Modern robot learning places large pretrained perception models at this first point of contact with the physical world. Vision-Language-Action (VLA) models ( $\pi_0$  [Black et al., 2024], OpenVLA [Kim et al., 2024], GR00T [Bjorck et al., 2025]) inherit visual encoders from Vision-Language Models (VLMs) such as SigLIP [Zhai et al., 2023] and DINOv2 [Oquab et al., 2023]. Modular systems chain open-vocabulary detectors (GroundingDINO [Liu et al., 2024b]), segmenters (SAM2 [Ravi et al., 2024]), and depth models (ZoeDepth [Bhat et al., 2023]) for grasp planning [Liu et al., 2024a, Ze et al., 2024]. World models (Tesseract [Zhen et al., 2025]) consume and generate depth and normal maps conditioned on actions. In each case, perception quality strongly influences downstream success.

**The unjustified backbone choice.** Despite this dependence, backbone selection is ad hoc. Li et al. [Li et al., 2025] confirmed that visual backbone choice dominates VLA policy success—more than architecture or training recipe—but evaluated entirely in simulation. OK-Robot [Liu et al., 2024a] showed that perception quality is the dominant bottleneck in real-world pick-and-place (success drops from 82% to 59% under clutter), but evaluated only one perception pipeline without comparing alternatives. The problem is not a lack of models—it is a lack of evidence about which model to use. Three deficiencies in current evaluation keep the choice unjustified. (i) **Sensor mismatch:** models are ranked on COCO [Lin et al., 2014] and ImageNet [Deng et al., 2009]—curated internet images at high resolution. Robots see through \$30 Intel RealSense D435 sensors at  $640 \times 480$  with structured-light depth noise, motion blur, and specular artifacts. No benchmark evaluates models on this imagery. (ii) **Task silos:** each task is evaluated on a different benchmark with different scenes, so per-task leaderboards cannot compose into a deployment recommendation. (iii) **No scene-state variation:** existing benchmarks capture static scenes. A robot’s scene changes as it acts—objects move, hands occlude, clutter shifts. No benchmark measures how perception degrades during manipulation and whether it recovers afterward.

**Evaluation methodology, not just a dataset.** The 2026 NeurIPS E&D track asks: *what claims does an evaluation support, under what assumptions?* We argue that answering “which perception backbone should a robot use?” requires **evaluation methodology** that existing benchmarks lack—not more data, but better measurement. Specifically, evaluation must: (a) create controlled scene-state variation so that performance differences are primarily attributable to scene state, not confounded by object or lighting changes; (b) measure properties beyond accuracy—temporal consistency, phase-transition robustness, geometric coherence—that determine whether a model *works in deployment*; and (c) stratify difficulty by perceptual ambiguity, not post-hoc geometry.

**RPX: methodology + dataset + toolkit.** We introduce **RPX**, which provides all three. The *three-phase capture protocol* (§3.1) mirrors the robot manipulation cycle (observe  $\rightarrow$  act  $\rightarrow$  verify), holding scene identity fixed while varying scene state. *Effort-Stratified Difficulty* (§3.2) derives difficulty from annotation effort—a direct signal of perceptual ambiguity. Three *deployment-readiness metrics* (§3.3)—Temporal Stability (TS), State-Transition Robustness (STR), and Geometric Coherence (SGC)—compose into an Embodied Deployment Score (EDS, §3.4) that ranks models for real-world use. These are instantiated on a 75,000-frame real-world RGB-D corpus across 100 scenes and  $\sim 70$  object categories (§4). All five tasks—depth, detection/grounding, tracking, scene QA, spatial QA—are evaluated on the *same* scenes and objects, described in crowd-sourced natural language [P et al., 2023], with a pip-installable toolkit (§5) that enables reproducible bring-your-own-model evaluation.

**Human proxy for robot perception.** A human carries the exact sensor rig a mobile robot would use—D435 (RGB-D at  $640 \times 480$ ) + T265 (stereo tracking) + GoPro (egocentric)—through 100 real-world scenes: kitchens, offices, outdoor paths, staircases, day and night, reflective and transparent surfaces. This is not a limitation but a deliberate design: the learning-from-humans paradigm [P et al.,

2026, Allu\* et al., 2025, Khazatsky et al., 2024, Kareer et al., 2024] already uses human-captured video as robot training data. If pretrained models fail on human-captured RealSense imagery, they fail on the data robots are trained on. Paired single-object and multi-object scenes enable controlled comparison of text-prompted vs. image-prompted perception under real-world conditions.

**Key findings.** Evaluating **TODO: N** models across five tasks, we find: **TODO: REPLACE with 2–3 concrete findings. Candidates:** (i) static accuracy is weakly correlated with deployment robustness ( $\rho = X$ ); (ii) model rankings on Hard scenes diverge from Easy ( $\tau = X$ ); (iii) VLM spatial QA accuracy drops by  $X\%$  during interaction, directly implicating VLA backbone robustness; (iv) EDS predicts downstream grasp success ( $\rho = X$ ).

## 81 Contributions.

- 82 1. **An evaluation methodology for deployment-ready robot perception.** Three-phase protocol +  
83 ESD + task-specific deployment-readiness metrics (TS, STR, SGC) + EDS—a methodological  
84 framework that can be applied to future datasets and tasks beyond **RPX**.
- 85 2. **The RPX dataset and multi-task benchmark.** 75,000-frame real-world RGB-D corpus across  
86 100 scenes and  $\sim 70$  object categories, with five perception tasks (depth, detection/grounding,  
87 tracking, scene QA, spatial QA) evaluated on the *same* scenes and objects—enabling the first  
88 cross-task deployment recommendations on shared real-world scenes.
- 89 3. **Empirical findings.** **TODO: 2–3 sentence findings summary, including the named finding.**
- 90 4. **Self-sustaining community toolkit.** rpx-benchmark: pip-installable, bring-your-own-model,  
91 with plugin registries for new tasks and metrics. 138 CI-verified tests.

92 **Scope.** **RPX** tests the claim that pretrained perception backbones differ in deployment robustness in  
93 ways not predictable from standard accuracy alone. It evaluates perception at the output level (depth  
94 maps, boxes, QA answers); for VLAs that consume encoder features, output-level metrics serve as  
95 proxies for feature quality (justified in §3.4, validated in §6.7). Assumptions and scope limitations  
96 are detailed in Appendix M.

## 97 2 Related Work

98 **RPX** addresses a gap at the intersection of six bodies of work: foundation models in robot learning  
99 stacks, vision-language-action (VLA) models and world models, robot perception datasets, embodied  
100 benchmarks, robustness and difficulty stratification, and in-context visual prompting for robotics.  
101 Table 1 provides a systematic comparison across 14 dimensions.

102 **Foundation models as perception substrates for robot learning.** Large pretrained vision models  
103 now occupy three distinct roles in robot manipulation. *As frozen visual backbones for policy learning*,  
104 DINOv2 [Oquab et al., 2023] and SigLIP [Zhai et al., 2023] provide the visual features that VLA  
105 models (OpenVLA [Kim et al., 2024],  $\pi_0$  [Black et al., 2024], GR00T N1 [Bjorck et al., 2025])  
106 consume to predict actions. The choice of visual encoder strongly shapes what the policy “sees,”  
107 yet it is typically inherited from the underlying VLM (e.g., PaliGemma  $\rightarrow$  SigLIP in  $\pi_0$ ) without  
108 independent evaluation on robot-relevant scenes. *As perception preprocessors in modular systems*,  
109 GroundingDINO [Liu et al., 2024b], SAM2 [Ravi et al., 2024], and monocular depth models feed  
110 signals into grasp planners, object trackers, and navigation stacks. OK-Robot [Liu et al., 2024a] is  
111 a paradigmatic example: it chains Lang-SAM, AnyGrasp, and a navigation planner for zero-shot  
112 pick-and-place in 10 real homes. Its analysis reveals that *perception quality is the primary bottleneck*:  
113 success drops from 82% in clean environments to 59% in cluttered ones, with perception failures  
114 (missed detections, poor segmentation) identified as the dominant cause [Liu et al., 2024a]—yet  
115 neither OK-Robot nor any other deployed system systematically compares the available perception  
116 backbones. *As pseudo-ground-truth generators*, ZoeDepth [Bhat et al., 2023] creates the 3D point  
117 clouds that 3D Diffusion Policy [Ze et al., 2024] trains on; ZoeDepth is also used to augment GR-  
118 1/GR00T fine-tuning with synthetic depth for embodiments lacking depth sensors. Inaccuracies in  
119 these pseudo-labels can propagate into the learned policy.

120 SceneReplica [Khargonkar et al., 2024] benchmarks real-world pick-and-place with swappable per-  
121 ception modules and confirms that perception dominates end-to-end success, but evaluates only 20

tabletop scenes without temporal dynamics or multi-task coverage. Open-X Embodiment [Collaboration, 2024] aggregates demonstration data at scale but focuses on action learning, not perception ranking. **RPX** is, to our knowledge, the first benchmark whose explicit purpose is to rank perception backbones across multiple tasks on the same real-world scenes.

**VLAs, world models, and the backbone selection problem.** The recent surge in VLA models—from RT-1/RT-2 [Brohan et al., 2023] through Octo [Ghosh et al., 2024] to  $\pi_0$  [Black et al., 2024], OpenVLA [Kim et al., 2024], and GR00T [Bjorck et al., 2025]—and embodied world models [Zhen et al., 2025] has made the visual backbone choice more consequential than ever. Li et al. [Li et al., 2025] directly study this in “What matters in building VLAs for generalist robots” (Nature Machine Intelligence): they systematically compare DINOv2, SigLIP, CLIP, and fusion strategies as VLA backbones and find that *backbone choice is the dominant factor in policy success*—more important than architecture or training recipe. However, their evaluation is conducted entirely in simulation (SimplerEnv, CALVIN), where the structured-light depth noise, motion blur, specular reflections, and hand occlusion dynamics of real deployment are absent. Crucially, because they evaluate end-to-end policy success, they *cannot decompose* whether a failure is due to perception or policy—a confound that perception-level evaluation resolves.

Embodied world models (Tesseract [Zhen et al., 2025], GWM [Lu et al., 2025a]) predict future RGB, depth, and normal frames conditioned on actions. Reward models increasingly rely on visual perception: VIP [Ma et al., 2023b] pre-trains visual reward functions from human video; ReWiND [Zhang et al., 2025] uses language-conditioned VLM rewards for task adaptation; Robometer [Liang et al., 2026] scales reward models across 1M+ trajectories with video-language understanding. These models consume perception signals (depth, segmentation) as training inputs and generate them as outputs; their quality is bounded by the fidelity of the underlying perception. Yet no existing benchmark evaluates whether these perception signals are *robust enough* under the deployment conditions the downstream systems will encounter. **RPX** provides exactly this evaluation: if ZoeDepth’s metric accuracy degrades by  $X\%$  during human interaction (as measured by STR), any world model or policy trained on ZoeDepth pseudo-labels inherits that degradation.

**Robot perception datasets.** Existing datasets each cover a fragment of the evaluation space **RPX** targets. **Object pose:** YCB-Video [Xiang et al., 2018] and BOP [Hodaň et al., 2024] provide 6-DoF pose for known objects with static cameras; SenseShift6D [Han et al., 2025] adds sensor-variation robustness but covers only 5 objects. **Segmentation:** OCID [Suchi et al., 2019] captures cluttered tabletops but has no temporal dynamics or camera pose; RISEg [Qian et al., 2024] uses robot interaction for segmentation without comparative backbone evaluation. **Reconstruction:** ScanNet [Dai et al., 2017], HM3D [Ramakrishnan et al., 2021], and EmbodiedScan [Wang et al., 2024] provide multi-modal 3D data but target reconstruction/navigation, not manipulation perception, and have no phase structure. **Human activity:** EPIC-Kitchens [Damen et al., 2018] and RH20T [Fang et al., 2023] contain interaction video but lack the instance masks, metric depth, and camera pose required for perception evaluation. **Robot-relevant objects:** FewSOL [P et al., 2023] and NIDS-Net [Lu et al., 2025b] study recognition on daily objects but not multi-task evaluation. **Large-scale robot data:** DROID [Khazatsky et al., 2024] aggregates 76K real-world manipulation trajectories across 564 scenes with multi-camera views, but targets policy learning rather than perception ranking—it provides no per-frame perception ground truth (depth, masks) or deployment-readiness metrics.

*No prior dataset evaluates multiple perception tasks on the same scenes and the same objects*—the confound that prevents per-task leaderboards from composing into a deployment recommendation.

**Embodied and multi-task benchmarks.** AI2-THOR [Kolve et al., 2017], Habitat [Savva et al., 2019], HomeRobot [Yenamandra et al., 2023], BEHAVIOR-1K [Li et al., 2023], and BEHAVIOR Vision Suite [Ge et al., 2024] enable multi-task evaluation in simulation—the latter with customisable scene generation—but cannot replicate real-world sensing (structured-light noise, specular reflections, motion blur). EmbodiedBench [Yang et al., 2025] benchmarks MLLMs as embodied agents but in simulation with action-level evaluation. RoboSense [Kong et al., 2025] tests sensor robustness for driving/navigation, not manipulation. WildCross [Knights et al., 2026] benchmarks outdoor depth and place recognition, complementary to **RPX**’s indoor manipulation focus. SQA3D [Ma et al., 2023a] and 3D-VisTA [Zhu et al., 2023] benchmark situated spatial reasoning on curated static 3D scenes but without phase transitions or embodied sensor dynamics. RoboVQA [Sermanet et al., 2023] asks long-horizon reasoning questions on robot-relevant real images but evaluates VLMs as planners,

not as perception backbones—and does not measure robustness to scene-state changes. **RPX** jointly evaluates multiple manipulation-relevant perception tasks under controlled phase-transition conditions on real-world scenes—a combination not provided by any benchmark in Table 1.

**Robustness and difficulty stratification.** Vision robustness benchmarks [Hendrycks and Dietterich, 2019, Taori et al., 2020] evaluate distributional corruptions but not the scene-state shifts that embodied manipulation induces. *Temporal consistency* has emerged as a model-level concern (Video Depth Anything [Chen et al., 2025], ChronoDepth [Shao et al., 2025], PTC-Depth [Li et al., 2026]), but these are model contributions, not benchmark contributions comparing across models. On the benchmark side, Spring [Mehl et al., 2023] provides high-resolution real-world sequences for depth and flow evaluation, and MOSE [Ding et al., 2023] stress-tests video object segmentation under complex occlusion dynamics—the closest tracking benchmark to **RPX**’s interaction phase. However, neither provides the controlled phase structure or deployment-readiness metrics that **RPX** introduces. *Difficulty* has been stratified by object size [Lin et al., 2014], occlusion [Hodaň et al., 2024], or viewpoint [Xiang et al., 2018]—post-hoc geometric splits. **RPX**’s ESD derives difficulty from *annotation-refinement effort*, grounding the split in perceptual ambiguity rather than geometry.

**Depth benchmarks.** The dominant depth benchmarks—NYU Depth V2 and KITTI—use static indoor/driving imagery with canonical viewpoints and no scene-state variation. VOID [Wong et al., 2020] adds camera motion (visual-inertial odometry) to depth evaluation but targets depth completion, not monocular estimation for manipulation. **RPX** evaluates monocular depth under the motion and occlusion dynamics of robot manipulation, with metrics (TS, SGC) that existing depth benchmarks do not compute.

**In-context visual prompting for robot perception.** A growing body of work uses *visual exemplars* rather than text to prompt perception models. DE-ViT [Zhang et al., 2024] performs few-shot detection from reference images without fine-tuning—“a basic skill for a robot in open environments.” NIDS-Net [Lu et al., 2025b] detects and segments novel instances from a few examples. DetPO [Ge et al., 2026] optimises in-context detection prompts for MLLMs. KUDA [Cheng et al., 2025] unifies visual prompting with keypoint-based manipulation. In VLA research, RICL [Sridhar et al., 2025] injects in-context adaptability into VLAs post-training. **RPX** supports this evaluation mode because it contains the *same objects* in both isolated single-object scenes (suitable as reference images) and cluttered multi-object scenes (the query). This enables a direct comparison: *does a text prompt (“red plastic bottle”) or a visual prompt (a reference photo of the bottle) produce more robust detection under clutter and interaction?* We are not aware of any benchmark that provides paired isolated/cluttered images of the same objects with shared annotations.

**The gap **RPX** fills.** Six properties jointly distinguish **RPX** from all prior work: (i) the same physical scenes across three controlled phase transitions; (ii) the same daily objects evaluated across all perception tasks simultaneously; (iii) 6-DoF hardware camera pose for temporal stability and phase-transition metrics; (iv) human interaction as a first-class experimental condition; (v) deployment-readiness metrics (TS, STR, SGC) and difficulty stratification (ESD) that go beyond accuracy; and (vi) paired single-object/multi-object scenes enabling in-context visual prompting evaluation. No existing benchmark provides all six (Table 1).

### 3 Evaluation Methodology

**RPX**’s central claim is that standard accuracy on curated benchmarks does not predict perception reliability under deployment. To test this, we design an evaluation methodology with three components: a *three-phase capture protocol* that creates controlled scene-state variation (§3.1), *Effort-Stratified Difficulty* that grounds difficulty in annotation effort rather than geometric heuristics (§3.2), and *deployment-readiness metrics* that measure properties invisible to static accuracy (§3.3). These compose into a single *Embodied Deployment Score* (§3.4) that ranks models for real-world use. The **RPX** dataset (§4) instantiates this methodology; the toolkit (§5) makes it reproducible and extensible.

#### 3.1 Three-Phase Scene Protocol

A deployed robot typically cycles through three perceptual states during a manipulation episode—*observe* → *act* → *verify*. We mirror this cycle:

Table 1: Systematic comparison of **RPX** with related datasets and benchmarks. ✓ = supported; ○ = partial; ✗ = not supported. Rows above the mid-rule are *dataset properties*; rows below are *evaluation methodology* features—a distinction we maintain to avoid conflating data contributions with measurement contributions. <sup>†</sup>Core datasets or task categories (not spatial scenes). <sup>‡</sup>Sequences or trials (not frames). <sup>§</sup>RoboVLMs is a methods paper, not a dataset; included for positioning only. DROID [Khazatsky et al., 2024] is a policy-learning dataset, not a perception benchmark, but included as the closest large-scale real-world comparison.

|                                   | <b>RPX</b> | RoboVLMs <sup>§</sup> | DROID | ScanNet | YCB-V | BOP            | OCID | RH20T             | FewSQL | EmbodiedScan | OK-Robot         | SceneReplica     | HomeRobot | EpiC-K   | WildCross |
|-----------------------------------|------------|-----------------------|-------|---------|-------|----------------|------|-------------------|--------|--------------|------------------|------------------|-----------|----------|-----------|
| Real-world capture                | ✓          | ✗                     | ✓     | ✓       | ✓     | ✓              | ✓    | ✓                 | ✓      | ✓            | ✓                | ✓                | ✗         | ✓        | ✓         |
| Phase structure (3-phase)         | ✓          | ✗                     | ✗     | ✗       | ✗     | ✗              | ✗    | ○                 | ✗      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| Multi-task GT on shared scenes    | ✓          | ✗                     | ✗     | ○       | ✗     | ✗              | ✗    | ○                 | ✗      | ○            | ✗                | ✗                | ✗         | ✗        | ○         |
| Same objects across all tasks     | ✓          | ✗                     | ✗     | ✗       | ✓     | ✓              | ✓    | ✗                 | ✓      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| Instance masks                    | ✓          | ✗                     | ✗     | ○       | ✓     | ✓              | ✓    | ✗                 | ✓      | ✓            | ✗                | ✗                | ✗         | ✗        | ✗         |
| Metric depth GT                   | ✓          | ✗                     | ○     | ✓       | ✓     | ✓              | ✓    | ✓                 | ✗      | ✓            | ✗                | ○                | ✗         | ✗        | ✓         |
| 6-DoF camera pose GT              | ✓          | ✗                     | ○     | ✓       | ○     | ✓              | ✓    | ✓                 | ✗      | ✓            | ✗                | ○                | ✗         | ✗        | ✓         |
| Language / grounding GT           | ✓          | ○                     | ✓     | ✗       | ✗     | ✗              | ✗    | ✓                 | ✓      | ✓            | ✓                | ✗                | ✗         | ○        | ✗         |
| Human interaction phase           | ✓          | ○                     | ✗     | ✗       | ✗     | ✗              | ✗    | ✗                 | ✗      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| Mobile-robot view (human proxy)   | ✓          | ✗                     | ✓     | ✓       | ✗     | ✗              | ✗    | ○                 | ✗      | ✓            | ✗                | ✗                | ✗         | ✓        | ✓         |
| Deployment-readiness metrics      | ✓          | ✗                     | ✗     | ✗       | ✗     | ✗              | ✗    | ✗                 | ✗      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| In-context visual prompt eval     | ✓          | ✗                     | ✗     | ✗       | ✗     | ✗              | ✗    | ✗                 | ○      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| Paired isolated / cluttered obj   | ✓          | ✗                     | ✗     | ✗       | ✗     | ✗              | ✗    | ✗                 | ○      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| Community-extensible toolkit      | ✓          | ✓                     | ✗     | ✗       | ✗     | ○              | ✗    | ✗                 | ✗      | ✗            | ○                | ○                | ○         | ✗        | ✗         |
| Perception-level backbone ranking | ✓          | ○                     | ✗     | ✗       | ✗     | ✗              | ✗    | ✗                 | ✗      | ✗            | ✗                | ✗                | ✗         | ✗        | ✗         |
| # Scenes                          | 100        | —                     | 564   | 1.5K    | 92    | 7 <sup>†</sup> | 96   | 147 <sup>†</sup>  | 336    | 5K           | 10               | 20               | sim       | 45       | 19        |
| # Frames                          | 75K        | —                     | 350K  | 2.5M    | 134K  | var.           | 2.4K | 110K <sup>‡</sup> | 3K     | 1M           | 171 <sup>‡</sup> | 100 <sup>‡</sup> | sim       | 20M      | 476K      |
| Eval. focus                       | deploy.    | policy                | imit. | recon.  | pose  | pose           | seg. | imit.             | recog. | 3D           | manip.           | manip.           | manip.    | activity | depth     |

228 **(1) Clutter** ( $\mathcal{P}_{\text{clu}}$ ): 360° walkaround of a densely arranged scene—the most frequent operating  
229 state [Liu et al., 2024a, Khazatsky et al., 2024].

230 **(2) Interaction** ( $\mathcal{P}_{\text{int}}$ ): a human rearranges the scene while D435 and GoPro capture simultaneous  
231 exo/ego views—closely matching the data that learning-from-human pipelines [P et al., 2026, Allu\*  
232 et al., 2025, Nair et al., 2022, Wang et al., 2023, Bharadhwaj et al., 2023, Kareer et al., 2024, Hoque  
233 et al., 2025, Khazatsky et al., 2024] train on. Perception failures here corrupt the demonstration data  
234 itself.

235 **(3) Clean** ( $\mathcal{P}_{\text{cln}}$ ): second walkaround of the reorganised scene—a within-scene control that tests  
236 post-manipulation verification.

237 Background, objects, lighting, and sensors are held fixed across phases. Camera trajectories are  
238 not identical (each walkaround follows a natural path), but the 360° coverage and ESD’s per-  
239 frame aggregation mitigate viewpoint confounds. Performance differences are therefore *primarily*  
240 *attributable to scene state*—enabling controlled real-world measurement of how perception degrades  
241 during manipulation and recovers afterward. Each phase yields ~250 frames (~75K total across 100  
242 scenes). The full deployment-grounded justification for exactly three phases is in Appendix A.

### 243 3.2 Effort-Stratified Difficulty (ESD)

244 Standard benchmarks stratify difficulty by 1–3 features within a single modality: KITTI [Geiger  
245 et al., 2012] uses bbox height, occlusion, and truncation; Waymo [Sun et al., 2020] uses LiDAR  
246 point count and a human flag; TARGO [Xia et al., 2024] uses occlusion level alone. Most major  
247 benchmarks (nuScenes, BOP, ScanNet, OCID) define no explicit difficulty splits at all. **RPX** takes a  
248 different approach: difficulty is grounded in **27 features spanning 8 categories**—annotation effort,  
249 scene complexity, occlusion, depth quality, photometric–depth conflict, temporal stability, camera  
250 motion, and fisheye/stereo quality—fusing signals from RGB, depth, instance masks, 6-DoF camera  
251 pose, T265 fisheye stereo, and annotation process logs. The insight that annotation process encodes  
252 difficulty builds on prior work in NLP (annotator disagreement as a calibration signal [Nie et al.,  
253 2020]) and vision (human response time for visual search as a difficulty proxy [Ionescu et al., 2016]),  
254 but to our knowledge, RPX is the first perception benchmark to use annotation refinement effort as a  
255 difficulty feature. The three-phase protocol further enables *phase-conditional* difficulty: splits are  
256 assigned per (scene, phase), not per-scene—a stratification only possible because the same scene is  
257 captured across controlled state transitions.

**Feature extraction.** For each (scene  $s$ , phase  $p$ ), we compute 27 features  $\{f_i\}$  spanning eight categories: annotation effort (2:  $f_{\text{iter-mean}}, f_{\text{iter-max}}$ ), scene complexity (3:  $f_{\text{obj-mean}}, f_{\text{obj-std}}, f_{\text{obj-consist}}$ ), occlusion (3:  $f_{\text{occ-mean}}, f_{\text{occ-p90}}, f_{\text{occ-heavy}}$ ), depth quality (4:  $f_{\text{depth-invalid}}, f_{\text{depth-invalid-mask}}, f_{\text{depth-std}}, f_{\text{depth-std-mask}}$ ), photometric–depth conflict (2:  $f_{\text{specular}}, f_{\text{dark}}$ ), temporal annotation stability (3:  $f_{\text{area-cv}}, f_{\text{area-drop}}, f_{\text{vis-instability}}$ ), camera motion (5:  $f_{\text{trans-mean}}, f_{\text{trans-p90}}, f_{\text{rot-mean}}, f_{\text{rot-p90}}, f_{\text{jerk}}$ ), and fisheye/stereo quality (5:  $f_{\text{fisheye-dark}}, f_{\text{fisheye-bright}}, f_{\text{fisheye-sharpness}}, f_{\text{fisheye-corr}}, f_{\text{fisheye-texture}}$ ; set to 0 when the T265 fisheye stream is absent). Full definitions: Appendix B.

**RPX Difficulty Score.** Features are percentile-normalised to  $[0, 1]$  and combined via a convex sum that anchors the score on annotation effort—the property the methodology is named for:

$$\text{RPX-DS}(s, p) = \alpha \cdot \frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} \tilde{f}_i(s, p) + (1 - \alpha) \cdot \frac{1}{F - |\mathcal{E}|} \sum_{i \notin \mathcal{E}} \tilde{f}_i(s, p)$$

where  $\mathcal{E} = \{f_{\text{iter-mean}}, f_{\text{iter-max}}\}$  are the annotation-effort features and  $\alpha \in [0, 1]$  controls how strongly effort anchors the score. The default **effort\_stratified\_v1** uses  $\alpha = 0.25$ : each effort feature carries 12.5% of the score (25% combined), while the 25 perception features collectively drive 75%—a modest emphasis on effort that respects the methodology’s name without letting it dominate. A **uniform\_v1** fallback ( $\alpha \rightarrow 0$  rebased so all 27 features carry  $1/F$ ) is shipped as a sensitivity baseline. Both are fully reproducible from data alone—no model dependency, no calibration-model checkpoints needed. A refined **mi\_v1** sets weights proportional to mutual information between each feature and per-scene model failure rate (Table 2), where failure is per-scene accuracy below the median across all scenes for that model. All three coexist in the splits manifest; **mi\_v1** is reported as a refinement when calibration models are available.

**Calibration procedure.** Because **RPX** is evaluation-only (all 99 scenes are test, with no train/val split), MI calibration uses *model holdout* rather than scene holdout: (i) weights are computed from a calibration model set of **TODO: N<sub>cal</sub>** models held out of the headline results; (ii) ESD validation (§3.2) uses models *not* in the calibration set; (iii) **uniform\_v1** is always available without calibration data, providing a frozen baseline that future researchers can reproduce from the dataset alone.

**Stability.** On the released 297 (scene, phase) pairs, the tier assignment is highly stable to data resampling: 89.9% of pairs retain the same dominant tier across 1,000 80%-bootstrap subsamples (Appendix D). Stability under weight perturbation depends on which prior we use as the centre: under the **uniform\_v1** prior, 88.6% of pairs retain the same dominant tier across 1,000 Dirichlet-sampled weight vectors over the simplex, with median Kendall  $\tau = 0.685$  (IQR: 0.630–0.734) between perturbed and centre rankings. Switching the primary from **uniform\_v1** to **effort\_stratified\_v1** ( $\alpha = 0.25$ ) shifts the per-(scene, phase) tier of 146 of 297 pairs (49.2%); the corresponding scene-level rollup (§ “ESD splits” below) shifts 20 of 99 scenes (20%, Adjusted Rand Index 0.50). The effort prior’s main empirical effect is to neutralise the counter-intuitive phase-composition inversion seen under uniform: under uniform, the interaction phase concentrates in Easy (46/99) because hand occlusion reduces per-frame complexity; under **effort\_stratified** the interaction phase redistributes across Medium and Hard, matching annotators’ observation that interaction frames required more refinement effort.

Table 2: Top-6 ESD features (of 27 total) ranked by mutual information with model failure rate on the calibration set (val split, **TODO: N<sub>cal</sub>** models). The remaining 21 features receive non-zero but lower weights; full weight vector in Appendix B.

| Feature                         | Weight $w_i$   | Spearman $\rho$ |
|---------------------------------|----------------|-----------------|
| $f_{\text{occ-mean}}$           | <b>TODO: X</b> | <b>TODO: X</b>  |
| $f_{\text{iter-mean}}$          | <b>TODO: X</b> | <b>TODO: X</b>  |
| $f_{\text{area-cv}}$            | <b>TODO: X</b> | <b>TODO: X</b>  |
| $f_{\text{depth-invalid-mask}}$ | <b>TODO: X</b> | <b>TODO: X</b>  |
| $f_{\text{vis-instability}}$    | <b>TODO: X</b> | <b>TODO: X</b>  |
| $f_{\text{jerk}}$               | <b>TODO: X</b> | <b>TODO: X</b>  |

**ESD splits.** **RPX** ships splits at two granularities. Per-(scene, phase) tertiles assign each of the 297 entries to Easy / Medium / Hard at the 33/33/34 cut on RPX-DS—enabling phase-conditional

analysis (Appendix D). The benchmark-consumer deliverable is the **per-scene** rollup: each scene’s three phase scores are averaged into a single mean, scenes are sorted, and the 99 (will be 100) scenes are cut into  $\lfloor N/3 \rfloor$  Easy +  $\lfloor N/3 \rfloor$  Medium + remainder Hard. With  $N = 99$  this yields a 33/33/33 split; with  $N = 100$  it becomes the canonical 33/33/34. Aggregating before tertiling guarantees that all three phases of a scene land in the same split—a hard requirement of the state-transition robustness metric (§ 3.3). The 33/33/34 cut is a *reporting convention* for interpretability (Easy/Medium/Hard is canonical in benchmarks), not a data-discovered cluster structure: silhouette and Tibshirani gap statistic on the 27-feature matrix both prefer  $k = 2$  over  $k = 3$  (Appendix D). To surface the methodological uncertainty this implies, we publish a *consensus tier* alongside the primary effort\_stratified\_v1 tier—majority vote across 11 scoring methods spanning percentile aggregation (mean, median, max, top-3), dimensionality reduction (PCA), and clustering (k-means at  $k \in \{3, 5, 8\}$ , GMM, Ward, DBSCAN; the latter is reported as degenerate when no density modes are found). Across the 297 (scene, phase) pairs, 8.1% receive unanimous agreement across methods, 35.0% are split between 2 tiers, and 56.9% across all 3—quantifying that scoring choice matters and motivating the consensus reporting alongside any single primary method.

**Validation.** To verify ESD captures genuine difficulty and to break the circularity of using SAM2 for both annotation and evaluation, we measure Spearman  $\rho$  between RPX-DS and model failure rate across all *non-SAM2*, *non-calibration* models:  $\rho = \textcolor{red}{TODO : X}$  ( $p < 0.001$ ). The capture-phase signal in feature space is weak—the maximum Adjusted Rand Index between any clustering method and the (clutter, interaction, clean) phase index is 0.182 (Appendix D)—confirming that ESD captures perception difficulty rather than capture-protocol artifacts. Mahalanobis-distance outlier detection (§D) flags 12 of 297 pairs as multivariate outliers; the top entries (scene100.jsom.atrium interaction phase, scene74.ecsw.atriumStairs clean phase) coincide with known T265 tracking failures during capture, demonstrating that the methodology surfaces real data quality issues.

### 3.3 Deployment-Readiness Metrics

Standard accuracy measures what a model gets right on one frame. But a robot’s perception feeds a pipeline: depth  $\rightarrow$  point cloud  $\rightarrow$  grasp planner, or VLM  $\rightarrow$  spatial reasoning  $\rightarrow$  target selection. Failures in three properties — temporal consistency, phase-transition robustness, and geometric fidelity — propagate into planning and control even when per-frame accuracy is high. Not all metrics apply to all tasks; Table 3 specifies which metrics are computed for each decision.

Table 3: Metric applicability. STR applies universally; SGC requires dense depth + instance masks (D1 only). TS is defined but restricted to non-interaction phases in this release (see text).

| Decision         | STR | TS             | SGC |
|------------------|-----|----------------|-----|
| D1: Depth        | ✓   | ○ <sup>†</sup> | ✓   |
| D2: Det./Ground. | ✓   | ✗              | ✗   |
| D3: Tracking     | ✓   | ✗*             | ✗   |
| D4: Scene QA     | ✓   | ✗              | ✗   |
| D5: Spatial QA   | ✓   | ✗              | ✗   |

<sup>†</sup>TS reported on clutter/clean phases only (interaction-phase pose unreliable); excluded from EDS in this release.

\*Tracking temporal consistency is captured by MOTA/IDF1’s inherent sequence-level design; a separate TS metric would be redundant.

**Temporal Stability (TS).** Pose-compensated frame-to-frame consistency for dense predictions. For depth, TS requires warping adjacent frames into a shared coordinate system using 6-DoF camera pose:  $\text{TS}_{\text{depth}} = \mathbb{E}_t \left[ 1 - \frac{\|D_t - \text{warp}(D_{t+1}, \Delta \mathbf{T}_t)\|_1}{N_{\text{valid}}} \right]$ , where  $\text{warp}(D, \Delta \mathbf{T})$  reprojects depth map  $D$  via the relative transform  $\Delta \mathbf{T}_t$  and  $N_{\text{valid}}$  counts pixels valid in both frames. *Why it matters for robots:* low TS means the point cloud jitters frame-to-frame, causing grasp planners to oscillate between candidates — even when each individual frame’s depth is accurate.

**Current status.** TS requires accurate extrinsic camera pose. In this release, T265 VIO pose exhibited sufficient drift during interaction-phase captures (hand occlusion of IR emitters) that TS measurements on those sequences would conflate model flicker with pose error. We therefore **report TS only on clutter and clean phases** (where T265 tracking is reliable) and **exclude TS from the EDS composite**

in this work. TS remains defined in the toolkit and will be activated when higher-quality pose (e.g., from marker-based tracking or SLAM with loop closure) is available—an explicit extension point. Importantly, the remaining EDS factors (accuracy, STR, SGC, efficiency) do not depend on camera pose and are fully operational across all three phases.

**State-Transition Robustness (STR).** The performance change across phase boundaries:  $\Delta_{\text{int}} = S_{\text{int}} - S_{\text{clu}}$  (interaction drop);  $\Delta_{\text{rec}} = S_{\text{clu}} - S_{\text{int}}$  (recovery). STR is defined for *all* five tasks: it uses each task’s primary accuracy metric (AbsRel for D1, AP<sub>50</sub> for D2, MOTA for D3, accuracy for D4/D5). A model with high accuracy but large  $|\Delta_{\text{int}}|$  is unreliable: its quality depends on scene state, a variable the robot cannot control. *Why it matters for robots:* during a pick-and-place episode, the scene transitions from clutter (pre-grasp) through interaction (grasping) to clean (post-grasp verification). A depth model that degrades during interaction produces bad point clouds exactly when the robot’s hand is in the scene — the moment grasp refinement is most critical. A VLM that loses spatial accuracy during interaction will misidentify targets for the next action.

**Geometric Coherence (SGC).**  $\text{SGC} = \text{F-score}(\partial M_{\text{gt}}, \{x : |\nabla \hat{D}(x)| > \tau_{\text{sgc}}\})$ —alignment between *ground-truth* mask boundaries and predicted depth discontinuities, with  $\tau_{\text{sgc}} = 1.5 \text{ mm/px}$  (derived from D435 depth resolution of  $\sim 2 \text{ mm}$  at  $1 \text{ m}$  and pixel pitch of  $\sim 1.4 \text{ mm/px}$  at  $640 \times 480$  with  $\sim 70^\circ$  HFOV; sensitivity analysis in Appendix C). SGC tests whether the depth map is geometrically consistent with the scene’s object structure. *Why it matters for robots:* grasp planners segment the point cloud along depth discontinuities to identify graspable surfaces. If depth edges do not align with actual object boundaries, the planner may attempt to grasp across two objects or miss a thin object entirely — failures invisible to AbsRel but fatal for manipulation. Because SGC requires dense depth predictions and GT instance masks, it is restricted to D1.

### 3.4 Embodied Deployment Score (EDS)

The final output of **R<sub>PX</sub>** for each (model, task) pair is the **Embodied Deployment Score**—a single scalar that tells a roboticist how suitable a model is for real-world deployment. EDS is designed to correlate with manipulation success: a model that is accurate, robust to scene-state changes, temporally stable, and fast enough for closed-loop control should produce better downstream performance than one that fails on any of these axes. EDS is a multiplicative composite: each factor is  $\in [0, 1]$  and any single poor factor drags the score down:

$$\text{EDS} = S_{\text{overall}} \cdot \text{Rob} \cdot \text{Stab} \cdot \eta$$

**Accuracy** ( $S_{\text{overall}}$ ):  $S_{\text{overall}} = \frac{1}{3}(S_{\text{clu}} + S_{\text{int}} + S_{\text{clu}})$ , where each  $S_p = 0.25M_{\text{Easy}} + 0.35M_{\text{Med}} + 0.40M_{\text{Hard}}$  (ESD-weighted). For lower-is-better metrics (e.g., AbsRel), we use  $S_p = 1 - \min(M_p, 1)$  so that all scores are  $\in [0, 1]$  with higher = better.

**Robustness** (Rob):  $\text{Rob} = 1 - \min(|\Delta_{\text{int}}|, 1)$ , where  $\Delta_{\text{int}} = S_{\text{int}} - S_{\text{clu}}$ . Clamping to  $[0, 1]$  ensures the factor cannot go negative even for catastrophic drops.

**Stability** (Stab): Designed to incorporate TS for dense tasks; in this release,  $\text{Stab} = 1$  for all tasks because TS is restricted to non-interaction phases (see above). EDS therefore reduces to  $S_{\text{overall}} \cdot \text{Rob} \cdot \eta$  — accuracy, robustness, and efficiency. When reliable full-sequence pose becomes available, TS can be reactivated as a fourth factor without changing the framework.

**Efficiency** ( $\eta$ ):  $\eta = \min(\frac{t_{\text{budget}}}{t_{\text{lat,median}}}, 1)$ , where  $t_{\text{budget}}$  is the real-time frame budget (33 ms for 30 fps; configurable per deployment). Models faster than budget get  $\eta = 1$ ; slower models are penalised linearly.

**Decomposability.** The individual components ( $S_{\text{overall}}$ ,  $\Delta_{\text{int}}$ ,  $\Delta_{\text{rec}}$ , TS, SGC, latency, memory) are always reported separately (Table 6), so practitioners can weight dimensions according to their deployment priorities. The multiplicative formulation is a conservative design choice: any single failing dimension (e.g., poor robustness) prevents a high score, appropriate for safety-relevant deployment. EDS is not prescriptive—it provides a sensible default ranking; the toolkit supports alternative compositions (additive weighting, Pareto analysis), and we verify in §6 that main findings are robust to the scoring rule.

386 **Connection to the perception–planning–control pipeline.** Table 4 maps each **R****P****X** metric to  
 387 the downstream failure mode it is designed to predict. This grounding ensures that EDS captures  
 388 properties that matter for deployed robots, not just properties that are easy to measure.

Table 4: How each **R****P****X** metric connects to the robot pipeline. Each metric targets a specific downstream failure mode.

| Metric               | Perception property                    | Downstream failure if poor                                                                |
|----------------------|----------------------------------------|-------------------------------------------------------------------------------------------|
| $S_{\text{overall}}$ | Per-frame accuracy across difficulties | Wrong depth/detections $\rightarrow$ wrong grasps, wrong targets                          |
| STR                  | Robustness to scene-state change       | Perception degrades when robot’s hand enters scene $\rightarrow$ grasp refinement fails   |
| TS <sup>†</sup>      | Frame-to-frame consistency             | Point cloud jitter $\rightarrow$ grasp planner oscillates; VLA gets inconsistent features |
| SGC                  | Depth edges align with objects         | Grasp planned across object boundaries; thin objects missed                               |
| $\eta$               | Inference speed                        | Cannot close the perception–action loop at control rate                                   |

389 **From outputs to features.** **R****P****X** evaluates perception at the *output* level (depth maps, boxes,  
 390 answers), but many robot learning systems consume perception at the *feature* level (encoder activations  
 391 inside VLAs). We argue the methodology remains informative for feature-level selection because:  
 392 (i) TS and STR measure *representation stability*—a backbone with flickering outputs under small  
 393 viewpoint changes likely has features that do not stably encode geometry, regardless of which head is  
 394 attached; (ii) D4/D5 directly evaluate the VLMs (PaliGemma, Qwen2-VL, GPT-4o) whose encoders  
 395 VLAs inherit, testing scene understanding on robot-grade imagery; and (iii) modular systems (OK-  
 396 Robot, DP3, Tesseract) consume raw outputs directly, so output-level evaluation is not a proxy but  
 397 the direct signal. The downstream validation (§6.7) empirically tests this link.

## 388 4 The **R****P****X** Dataset

389 The methodology of §3 requires a dataset with four properties: (i) the same physical scenes captured  
 400 across controlled phase transitions, (ii) multiple perception modalities with shared ground truth,  
 401 (iii) 6-DoF camera pose for temporal metrics, and (iv) objects described in natural language for  
 402 grounding evaluation. We are not aware of any existing dataset that provides all four (Table 1). We  
 403 construct **R****P****X** to fill this gap.

404 **Sensor platform.** A hand-carried rig pairs an Intel RealSense **D435** (RGB-D, structured IR,  
 405 0.3–3 m) with a **T265** (stereo visual-inertial, 6-DoF pose at 200 Hz) and a head-mounted **Go-  
 406 Pro** during interaction. Human-carried capture is a deliberate proxy for embodied robot motion,  
 407 validated via motion-profile comparison with deployed platforms (Appendix G): median  
 408  $\Delta t = \text{TODO} : X$  cm/frame, median  $\Delta R = \text{TODO} : X^\circ$ /frame, within the envelope of Stretch  
 409 RE2, Unitree H1, and Fetch.

410 **Scale and diversity.** 99 real-world scenes (**TODO: X** indoor, **TODO: X** outdoor), spanning daylight  
 411 and night, reflective and transparent surfaces.<sup>1</sup>  $\sim 70$  daily object categories (**TODO: X** total physical  
 412 instances) following FewSOL [P et al., 2023]: real household items varying in material, surface, and  
 413 size.  $\sim 75,000$  evaluation frames across three phases at  $640 \times 480$ , 30 fps. **Modalities:** RGB, metric  
 414 depth, 6-DoF camera pose, instance masks, tracklets, per-object textual attributes, T265 fisheye stereo  
 415 pairs, and GoPro egocentric video during interaction.

416 **Objects and language annotations.** Per-object textual attributes (category, colour, material, shape)  
 417 use layperson vocabulary collected via Amazon Mechanical Turk for the FewSOL object set—  
 418 matching how real users will command robots. The same objects appear in both single-object  
 419 (isolated, AR-tagged) and multi-object (cluttered, three-phase) scenes, enabling in-context visual  
 420 prompting evaluation (§5).

<sup>1</sup>The collection target was 100 scenes; one scene was excluded from v1 due to an incomplete capture and is planned for v1.1.

Table 5: Five perception decisions. Each evaluates a signal consumed by deployed robot systems.

| #  | Decision          | Systems                                                     | Models                                                             | Metrics                       | Diagnostic           |
|----|-------------------|-------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------|----------------------|
| D1 | Depth model       | DP3 [Ze et al., 2024], GR00T [Bjorck et al., 2025]          | DA-V2, Video DA, Depth Pro, UniDepth, Metric3D, ZoeDepth           | AbsRel, RMSE, $\delta < 1.25$ | STR, TS, SGC         |
| D2 | Detector grounder | / OK-Robot [Liu et al., 2024a]                              | Text: GroundingDINO, OWL-ViT, YOLO-World. Visual: DE-ViT, NIDS-Net | AP <sub>50</sub> , Acc@0.5    | STR, text vs. visual |
| D3 | Tracker           | Pick-and-place, re-grasping                                 | SAM2 video, Cutie                                                  | MOTA, IDF1                    | ID switches/phase    |
| D4 | Scene VLM         | QA $\pi_0$ [Black et al., 2024], OpenVLA [Kim et al., 2024] | GPT-4o, Qwen2-VL, LLaVA, Molmo 2, PaliGemma                        | Fuzzy-match Acc               | STR on QA            |
| D5 | Spatial VLM       | QA Spatial manipulation                                     | Same as D4                                                         | Fuzzy-match Acc               | STR on spatial QA    |

**Ground truth.** *Instance masks:* SAM2 auto-mode + GroundingDINO proposals + iterative human refinement; refinement count  $k_j$  feeds ESD (§3.2). *Tracklets:* SAM2 video propagation with human-corrected keyframes. *Depth:* raw D435 structured-light; invalid pixels flagged, no inpainting. *Camera pose:* T265 VIO at 200 Hz, downsampled to 30 fps. Pose quality is reliable during clutter and clean phases (smooth walkaround motion) but degrades during interaction (hand occlusion of T265 IR emitters). TS is therefore restricted to non-interaction phases in this release (§3.3). *QA ground truth:* human-authored question templates instantiated deterministically from annotations (§5). Full annotation pipeline details and figures in Appendix E.

**Availability.** HuggingFace (<https://anonymous.4open.science/r/RPX>), CC BY 4.0. 5-scene reviewer sample (<4 GB); full dataset ~**TODO: X** GB. Croissant metadata (core + RAI fields) included. All 99 scenes are released for evaluation; **RPX** is a benchmark for models trained elsewhere, so no train/val split is provided. Difficulty stratification (Easy/Medium/Hard) is shipped at two granularities in the splits manifest: (i) `scene_splits.json`—a per-scene 33/33/33 (33/33/34 once the 100<sup>th</sup> scene lands) tier assignment, the primary deliverable for benchmark consumers, with all three phases of a scene held in the same tier to satisfy the state-transition robustness requirement; (ii) `phase_difficulty.json`—per-(scene, phase) structured metric with the primary effort\_stratified\_v1 tier, cross-method consensus tier, per-row weight-perturbation confidence flag, and an 8-d per-modality sub-score vector that preserves the multi-faceted nature of difficulty (§3.2, Appendix D).

## 5 Benchmark Tasks & Toolkit

Every robot learning pipeline makes implicit perception choices that are rarely justified. RoboVLMs [Li et al., 2025] confirmed that backbone selection dominates VLA success, yet evaluated only in simulation. **RPX** provides the real-world evaluation by framing the benchmark as **five perception decisions** (Table 5), all evaluated *zero-shot* on the same physical scenes and objects.

**D1: Monocular depth.** Predict a dense metric depth map from a single RGB frame (no scale alignment). Metric depth is the primary input for point-cloud-based grasping [Ze et al., 2024] and the pseudo-GT that world models [Zhen et al., 2025] train on. GT: D435 depth with validity mask.

**D2: Detection, grounding, and in-context visual recognition.** Localise target objects from a *text* prompt (category name or referring expression from questionnaire) or a *reference image* (from single-object scene). **RPX** supports both modes on the same physical objects, enabling a controlled

comparison: *which prompting mode degrades less across phases?* GT: bounding boxes from instance masks.

**D3: Multi-object tracking.** Maintain consistent object identities across frames. The interaction phase is the tracking stress test: hand occlusion causes ID switches—a primary failure mode in demonstration data. GT: SAM2 video-propagated tracklets with human-corrected keyframes; initialised with GT first-frame masks (isolating tracking from detection). To avoid circularity when evaluating SAM2 as a tracker, we report SAM2’s results alongside a note that its GT was generated by the same model family, and we verify that human corrections are sufficient to make the GT independent of any single model’s failure modes (Appendix E).

**D4: Scene understanding QA.** Given an RGB frame and a question (“What objects are on the table?”), produce a textual answer. VLAs are built on VLMs; if the VLM cannot answer correctly on D435 imagery, the VLA is likely to misidentify targets. Evaluation uses fuzzy matching (normalised string similarity  $\geq 0.8$  after lowercasing and article removal) to handle synonym and formatting variation. Examples: “a red bottle” matches “red bottle” (✓); “there are 3 items” matches “3 items” (✓); “the bottle is on the table” does not match “bottle” (✗, similarity  $< 0.8$ ). GT: human-authored question templates grounded in robotics operator needs (object identity, attribute verification, count, state change), instantiated deterministically per frame using questionnaire attributes and per-frame visibility masks. The same question has *different correct answers* across phases. Unlike static VQA benchmarks (VQAv2, GQA) or 3D-situated QA (SQA3D [Ma et al., 2023a]), **RPX** tests on robot-grade sensor imagery ( $640 \times 480$ , D435 colour) with phase-varying GT and crowd-sourced natural-language object descriptions. Template design and answer-key validation details are in Appendix E.2.

**D5: Spatial reasoning QA.** Given a spatial question (“What is to the left of the bottle?”), produce a textual answer. All spatial relations are defined in the *image plane*, matching how VLMs process images. Critically, spatial relations *change* across phases—making this, to our knowledge, the first perception benchmark with phase-varying spatial QA ground truth. GT: human-designed spatial-relation templates (left/right/above/below/near/far), instantiated deterministically from bounding-box geometry; templates were authored to reflect the spatial queries a robot operator or planner would issue.

**Toolkit.** `rpx-benchmark` (pip-installable, MIT) encapsulates the full pipeline: `rpx bench monocular_depth -hf-checkpoint my-model -split hard`. Three plugin registries (task, metric, model adapter) enable community extension without core changes. 138 CI-verified tests ensure reproducibility. Extended documentation in Appendix J.

## 6 Experiments

### 6.1 Setup

All models are evaluated **zero-shot**—no fine-tuning or adaptation to **RPX** objects or scenes. For detection/grounding, text prompts are crowd-sourced object descriptions from questionnaires (§4); visual prompts are reference images from single-object scenes. For tracking, SAM2 is initialised with GT first-frame masks, isolating tracking from detection. For QA, questions are generated from questionnaire attributes (D4) and bounding-box spatial layout (D5); answers are verified against per-frame annotations. Evaluations run on **TODO: hardware**. For each model we report parameter count (M), FLOPs (G) from a single forward pass at  $640 \times 480$ , peak GPU memory (GB), and median per-sample latency. All accuracy metrics are reported as mean  $\pm$  standard deviation across test scenes; latency is reported with 95% confidence intervals—enabling both statistical and hardware-agnostic comparison across backbones.

### 6.2 Main Results

Table 6 reports primary results across all five decisions.

Table 6: Main **RPX** results.  $S_{ov}$ : ESD-weighted overall score ( $S_{overall}$ , §3.4).  $\Delta_{int}$ : interaction drop.  $\Delta_{rec}$ : recovery. TS: temporal stability (clutter/clean phases only; excluded from EDS). Mem: peak GPU memory. Lat: avg. per-sample latency (95% CI). **Bold**: best per task. Underline: second.

| Decision                               | Model                | $S_{ov}$ | $\Delta_{int}$ | $\Delta_{rec}$ | TS†     | Params  | FLOPs   | Mem      | Lat      |
|----------------------------------------|----------------------|----------|----------------|----------------|---------|---------|---------|----------|----------|
| D1: Depth<br>(AbsRel↓)                 | DA-V2 Large          | TODO: X  | TODO: X        | TODO: X        | TODO: X | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | Video Depth Anything | TODO: X  | TODO: X        | TODO: X        | TODO: X | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | Depth Pro            | TODO: X  | TODO: X        | TODO: X        | TODO: X | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | UniDepth V2          | TODO: X  | TODO: X        | TODO: X        | TODO: X | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | Metric3D V2          | TODO: X  | TODO: X        | TODO: X        | TODO: X | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | ZoeDepth             | TODO: X  | TODO: X        | TODO: X        | TODO: X | TODO: M | TODO: G | TODO: GB | TODO: ms |
| D2: Det/Ground<br>(AP <sub>50</sub> ↑) | GroundingDINO (text) | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | OWL-ViT v2 (text)    | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | DE-ViT (visual)      | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | NIDS-Net (visual)    | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
| D3: Tracking<br>(MOTA↑)                | SAM2 video           | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | Cutie                | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
| D4: Scene QA<br>(Acc↑)                 | GPT-4o               | TODO: X  | TODO: X        | TODO: X        | –       | N/A     | N/A     | N/A      | TODO: ms |
|                                        | Qwen2-VL             | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | LLaVA                | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | Molmo 2              | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | PaliGemma            | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
| D5: Spatial QA<br>(Acc↑)               | GPT-4o               | TODO: X  | TODO: X        | TODO: X        | –       | N/A     | N/A     | N/A      | TODO: ms |
|                                        | Qwen2-VL             | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | LLaVA                | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | Molmo 2              | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |
|                                        | PaliGemma            | TODO: X  | TODO: X        | TODO: X        | –       | TODO: M | TODO: G | TODO: GB | TODO: ms |

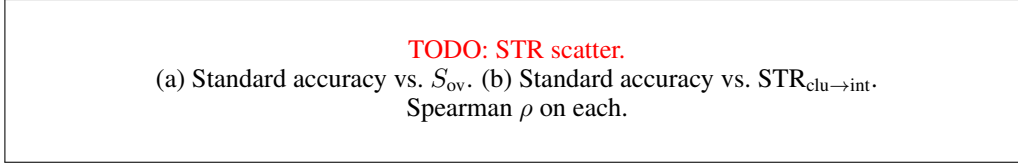


Figure 2: Standard accuracy vs. **RPX** deployment robustness.  $\rho = \text{TODO} : X$ : static rankings are **TODO**: poor predictors of deployment performance.

### 6.3 Does Static Accuracy Predict Deployment Robustness?

**TODO: REPLACE: 3–4 sentences naming specific models.**

### 6.4 Text vs. Visual Prompting

**TODO: REPLACE: Which mode degrades less? Key implications for pipeline design.**

### 6.5 VLM Scene Understanding Across Phases

**TODO: REPLACE: Key findings on VLM robustness. Name which VLM is most/least robust. Connect to VLA implications: “if PaliGemma’s spatial QA drops by X% during interaction,  $\pi_0$ ’s spatial instruction following is degraded by the same margin.”**

### 6.6 ESD Difficulty and Phase-Transition Analysis

**TODO: REPLACE: ESD ranking divergence ( $\tau$ ), phase-transition patterns, named models. Extended per-task breakdowns in Appendix F.**

### 6.7 Downstream Validation: Does EDS Predict Grasp Success?

A deployment-readiness score is useful only if it predicts downstream task success. The central validation question: *does a model’s EDS rank predict its contribution to manipulation success?* We test this by running all **TODO: N** evaluated depth models (ranked by EDS) through a grasping pipeline on **RPX** scenes and measuring rank-order correlation between EDS and grasp success rate. A positive result validates EDS as a deployment-relevant metric; a negative result would indicate that the proposed metrics do not capture what matters for manipulation.

TODO: Text vs visual prompt.  
Grouped bars: text AP vs visual AP, per phase.

Figure 3: Text vs. image-prompted detection across three phases.

TODO: QA accuracy by phase.  
Left: General QA (D4) per phase. Right: Spatial QA (D5) per phase.  
Show whether spatial QA drops more than general QA during interaction.

Figure 4: VLM QA accuracy across three phases. Spatial QA (D5) **TODO: drops more / less than** general QA (D4) during interaction, indicating **TODO: describe implication for VLA backbone robustness**.

516 **TODO: REPLACE: Describe pipeline (e.g. Contact-GraspNet on RPX depth predictions), number of**  
517 **trials, result. Key number: Spearman  $\rho$  between EDS and grasp success rate. Even a small experiment**  
518 **(5–10 trials per model) validates the entire scoring framework.**

519 **Independent validation: ranking divergence.** As a complementary check, we compare **RPX**  
520 **depth rankings with published NYU Depth V2 and KITTI rankings. Because the evaluation images,**  
521 **metrics, and protocols differ, rank divergence does not prove standard benchmarks are “wrong”—**  
522 **but systematic disagreement supports the claim that standard benchmarks alone are insufficient**  
523 **for deployment-specific model selection. TODO: REPLACE: Show Kendall  $\tau$  between RPX and**  
524 **NYU/KITTI rankings. Characterise which models change rank most.**

## 525 7 Conclusion

526 We presented an evaluation methodology for measuring the deployment readiness of foundation-  
527 model perception under embodied robot conditions, instantiated through **RPX**. The three-phase  
528 protocol enables controlled measurement of perception degradation and recovery under scene-state  
529 variation; ESD grounds difficulty in annotation effort; and deployment-readiness metrics (STR for all  
530 tasks; TS and SGC for depth) capture properties invisible to static accuracy. Together, these compose  
531 into the Embodied Deployment Score (EDS)—a single actionable ranking per model.

532 **TODO: REPLACE: 2–3 concrete findings with numbers. E.g.: “Evaluating N models, we find that (i)**  
533 **standard-benchmark rankings are weakly correlated with deployment robustness ( $\rho = X$ ); (ii) model**  
534 **rankings on Hard scenes diverge from Easy ( $\tau = X$ ); (iii) [named finding].”**

535 **Limitations. Sensor envelope:** indoor scenes within D435 range (0.3–3 m); results may not  
536 generalise to outdoor or long-range sensing. **Motion proxy:** human-carried capture approximates but  
537 does not replicate all robot motion profiles; validated via motion-statistics comparison with deployed  
538 platforms (§4, Appendix G), but platforms with non-humanoid kinematics (e.g., fixed-base arms) may  
539 exhibit different failure modes. **Pose quality:** T265 VIO pose is unreliable during interaction-phase  
540 captures (hand occlusion of IR emitters), restricting TS to clutter and clean phases and excluding it  
541 from EDS in this release. This is a data-collection limitation, not a methodological one: the TS metric  
542 and its EDS integration are fully defined and will activate when higher-quality pose is available.  
543 **ESD circularity:** annotation-effort features are derived from SAM2-based annotation; mitigated  
544 by validating ESD against non-SAM2 model failure rates on a held-out split (§3.2). **Scale:** 15 test  
545 scenes  $\times$  3 phases = 45 scene-phase pairs per ESD level ( $\sim 15$ ); we report standard deviations across  
546 scenes to characterise statistical power. **Metric scope:** TS and SGC are currently defined only for D1  
547 (depth); extending them to detection (box jitter) and segmentation (mask flicker) is an explicit toolkit  
548 extension point. **Output vs. feature evaluation:** **RPX** evaluates perception at the output level; for  
549 VLAs that consume encoder features rather than explicit outputs, **RPX**’s metrics serve as proxies for  
550 feature quality (§3.4), validated empirically in §6.7.

TODO: ESD violins + phase heatmap.  
Left: per-task violins. Right: models  $\times$  phases heatmap.

Figure 5: ESD separates models; phase transitions reveal robustness profiles.

TODO: EDS vs. grasp success.  
x-axis: EDS rank. y-axis: grasp success rate.  
Spearman  $\rho$  between EDS and grasp success.

Figure 6: EDS predicts downstream grasp success. Models ranked higher by RPX’s deployment-readiness score achieve higher grasp success rates, validating EDS as a deployment-relevant metric.

551 **Broader impact.** RPX evaluates perception on daily household objects; no personal data or faces  
552 are captured (IRB-exempt; consent obtained from interaction-phase participants). Improving percep-  
553 tion robustness enables more reliable manipulation with positive societal impact (assistive robotics,  
554 logistics). We acknowledge that more robust object detection could in principle be repurposed for  
555 surveillance; however, the models evaluated are already publicly available, and RPX adds evaluation  
556 methodology, not new detection capability.

557 **Community extensions.** We release data, annotations, splits, Croissant metadata, and the  
558 rpx-benchmark toolkit at <https://anonymous.4open.science/r/RPX>. The toolkit’s plugin  
559 architecture sustains community growth: new tasks, metrics, and models require no core changes.  
560 Released but not evaluated in this work: T265 fisheye stereo pairs and GoPro egocentric video  
561 captured during the interaction phase. The GoPro footage shows the same objects and manipulation  
562 events that the D435 pipeline annotates, providing a ready-made resource for the ego-exo perception  
563 research community [Nair et al., 2022, Kareer et al., 2024, Hoque et al., 2025]. Future work with  
564 temporal synchronisation (e.g., audio alignment or hardware timecode) would enable ego-exo consis-  
565 tency benchmarking—measuring whether models produce consistent outputs from both viewpoints  
566 simultaneously.

## 567 References

- 568 Mubashara Akhtar, Omar Benjelloun, Costanza Conforti, Joan Giner-Miguel, Nitisha Goel, Aibek  
569 Guha, Eitan Hess, Ferrante Jaïem, Michael Kuchnik, Quentin Lhoest, et al. Croissant: A metadata  
570 format for ML-ready datasets. In *DEEM Workshop at SIGMOD*, 2024.
- 571 Sai Haneesh Allu\*, Jishnu Jaykumar P\*, Ninad Khargonkar, Tyler Summers, Jian Yao, and Yu Xiang.  
572 Hrt1: One-shot human-to-robot trajectory transfer for mobile manipulation, 2025. URL <https://arxiv.org/abs/2510.21026>.
- 573
- 574 Shikhar Bahl, Abhinav Gupta, and Deepak Pathak. Human-to-robot imitation in the wild. In *RSS*,  
575 2022.
- 576 Homanga Bharadhwaj, Abhinav Gupta, Shubham Tulsiani, and Vikash Kumar. Zero-shot robot  
577 manipulation from passive human videos. *arXiv preprint arXiv:2302.02011*, 2023.
- 578 Homanga Bharadhwaj, Roozbeh Mottaghi, Abhinav Gupta, and Shubham Tulsiani. Track2act:  
579 Predicting point tracks from internet videos enables generalizable robot manipulation. In *ECCV*,  
580 2024.
- 581 Shariq Farooq Bhat, Reiner Birkel, Diana Wofk, Peter Wonka, and Matthias Müller. ZoeDepth:  
582 Zero-shot transfer by combining relative and metric depth. *arXiv preprint arXiv:2302.12288*, 2023.
- 583 Nils Bjorck et al. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv*  
584 *preprint*, 2025.

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, et al.  $\pi_0$ : A vision-language-action flow model for general robot control. *arXiv preprint arXiv:2410.24164*, 2024.

Anthony Brohan et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *CoRL*, 2023.

Chi-Lam Cheang et al. Gr-2: A generative video-language-action model with web-scale knowledge for robot manipulation. *arXiv preprint arXiv:2410.06158*, 2024.

Sili Chen, Hengkai Guo, Shengnan Zhu, Feihu Zhang, Zilong Huang, Jiashi Feng, and Bingyi Kang. Video Depth Anything: Consistent depth estimation for super-long videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. Highlight.

Chuyue Cheng et al. Kuda: Keypoint-based unified visual prompting for manipulation. *arXiv preprint arXiv:2503.10546*, 2025.

Open X-Embodiment Collaboration. Open X-embodiment: Robotic learning datasets and RT-X models. In *ICRA*, 2024.

Angela Dai, Angel X Chang, Manolis Savva, Maciej Halber, Thomas Funkhouser, and Matthias Nießner. ScanNet: Richly-annotated 3D reconstructions of indoor scenes. In *CVPR*, 2017.

Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, et al. Scaling egocentric vision: The EPIC-Kitchens dataset. In *ECCV*, 2018.

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009.

Henghui Ding, Chang Liu, Shuting He, Xudong Jiang, Philip H.S. Torr, and Song Bai. MOSE: A new dataset for video object segmentation in complex scenes. In *ICCV*, 2023.

Hao-Shu Fang, Hongjie Fang, Zhenyu Tang, Jirong Liu, Junbo Wang, Haoyi Zhu, and Cewu Lu. RH20T: A robotic dataset for learning diverse skills in one-shot. *arXiv preprint arXiv:2307.00595*, 2023.

Yixiao Ge et al. Detpo: In-context learning with multi-modal llms for few-shot object detection. *arXiv preprint arXiv:2603.23455*, 2026.

Yunhao Ge, Yihe Tang, Jiashu Xu, et al. BEHAVIOR vision suite: Customizable dataset generation via simulation. In *CVPR (Highlight)*, 2024.

Andreas Geiger, Philip Lenz, and Raquel Urtasun. Are we ready for autonomous driving? The KITTI Vision Benchmark Suite. In *CVPR*, 2012.

Dibya Ghosh, Homer Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan You, et al. Octo: An open-source generalist robot policy. In *RSS*, 2024.

Kristen Grauman, Andrew Westbury, Eugene Byrne, et al. Ego4d: Around the world in 3,000 hours of egocentric video. In *CVPR*, 2022.

Yegyu Han, Taegyoon Yoon, Dayeon Woo, Sojeong Kim, and Hyung-Sin Kim. Senseshift6d: Multimodal rgb-d benchmarking for robust 6d pose estimation across environment and sensor variations. *arXiv preprint arXiv:2507.05751*, 2025.

Xiaofei He and Partha Niyogi. Locality preserving projections. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2003.

Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *ICLR*, 2019.

629 Tomáš Hodaň, Martin Sundermeyer, Yann Labbe, Eric Brachmann, Vincent Lepetit, Carsten Rother,  
630 and Jiří Matas. BOP challenge 2023 on detection, segmentation and pose estimation of seen and  
631 unseen rigid objects. *arXiv preprint arXiv:2403.09799*, 2024.

632 Ryan Hoque, Peide Huang, David J. Yoon, Mouli Sivapurapu, and Jian Zhang. Egodex: Learning  
633 dexterous manipulation from large-scale egocentric video. *arXiv preprint arXiv:2505.11709*, 2025.

634 Radu Tudor Ionescu, Bogdan Alexe, Marius Leordeanu, Marius Popescu, Dim P. Papadopoulos, and  
635 Vittorio Ferrari. How hard can it be? Estimating the difficulty of visual search in an image. In  
636 *CVPR*, 2016.

637 Simar Kareer, Dhruv Patel, Ryan Punamiya, Pranay Mathur, Shuo Cheng, Chen Wang, Judy Hoffman,  
638 and Danfei Xu. Egomimic: Scaling imitation learning via egocentric video. *arXiv preprint*  
639 *arXiv:2410.24221*, 2024.

640 Ninad Khargonkar, Sai Haneesh Allu, Yangxiao Lu, Jishnu Jaykumar P, Balakrishnan Prabhakaran,  
641 and Yu Xiang. Scenereplica: Benchmarking real-world robot manipulation by creating replicable  
642 scenes. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages  
643 8258–8264, 2024. doi: 10.1109/ICRA57147.2024.10610180.

644 Alexander Khazatsky, Karl Pertsch, Suraj Nair, et al. Droid: A large-scale in-the-wild robot manipu-  
645 lation dataset. In *RSS*, 2024.

646 Moo Jin Kim et al. Openvla: An open-source vision-language-action model. *CoRL*, 2024.

647 Joshua Knights, Joseph Reid, Kaushik Roy, David Hall, Mark Cox, and Peyman Moghadam. Wild-  
648 cross: A cross-modal large scale benchmark for place recognition and metric depth estimation in  
649 natural environments. In *ICRA*, 2026.

650 Eric Kolve, Roozbeh Mottaghi, Winson Han, Eli VanderBilt, Luca Weihs, Alvaro Herrasti, Daniel  
651 Gordon, Yuke Zhu, Abhinav Gupta, and Ali Farhadi. AI2-THOR: An interactive 3d environment  
652 for visual ai. In *arXiv preprint arXiv:1712.05474*, 2017.

653 Haoran Kong et al. Robosense: Large-scale dataset and benchmark for egocentric robot perception  
654 and navigation in crowded and unstructured environments. *NeurIPS Datasets and Benchmarks*,  
655 2025.

656 Chengshu Li, Ruohan Zhang, Josiah Wong, Aaron Gokaslan, Sanjana Campose, Sanjana Srivastava,  
657 Adrien Herrmann, Shyamal Maher, Fei Xia, Roberto Martín Moro, et al. BEHAVIOR-1K: A  
658 benchmark for embodied AI with 1000 everyday activities and realistic simulation. In *CoRL*, 2023.

659 Xinghang Li, Peiyan Li, Long Qian, Minghuan Liu, Dong Wang, Jirong Liu, Bingyi Kang, Xiao Ma,  
660 Xinlong Wang, Di Guo, Tao Kong, Hanbo Zhang, and Huaping Liu. What matters in building  
661 vision-language-action models for generalist robots. *Nature Machine Intelligence*, 2025.

662 Xuanhe Li et al. Ptc-depth: Pose-refined monocular depth estimation with temporal consistency.  
663 *CVPR*, 2026.

664 Anthony Liang, Yigit Korkmaz, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sidhant Kaushik,  
665 Aditya Shah, Alex S. Huang, Luke Zettlemoyer, Dieter Fox, Yu Xiang, Anqi Li, Andreea Bobu,  
666 Abhishek Gupta, Stephen Tu, and Erdem Biyik. Robometer: Scaling general-purpose robotic  
667 reward models via trajectory comparisons. *arXiv preprint arXiv:2603.02115*, 2026.

668 Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr  
669 Dollár, and C Lawrence Zitnick. Microsoft COCO: Common objects in context. In *ECCV*, 2014.

670 Peiqi Liu, Yaswanth Orru, Jay Vakil, Chris Paxton, Nur Muhammad Mahi Shafiullah, and Lerrel  
671 Pinto. Ok-robot: What really matters in integrating open-knowledge models for robotics. In *RSS*,  
672 2024a.

673 Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan  
674 Li, Jianwei Yang, Hang Su, et al. Grounding dino: Marrying dino with grounded pre-training for  
675 open-set object detection. In *European Conference on Computer Vision*, pages 38–55. Springer,  
676 2024b.

677 Guanxing Lu, Baoxiong Jia, Puhao Li, Yixin Chen, Ziwei Wang, Yansong Tang, and Siyuan Huang.  
678 Gwm: Towards scalable gaussian world models for robotic manipulation. *ICCV*, 2025a.

679 Yangxiao Lu, Jishnu Jaykumar P, Yunhui Guo, Nicholas Ruozzi, and Yu Xiang. Adapting pre-trained  
680 vision models for novel instance detection and segmentation. In *IROS*, 2025b.

681 Xiaojian Ma, Silong Yong, Zilong Zheng, Qing Li, Yitao Liang, Song-Chun Zhu, and Siyuan Huang.  
682 SQA3D: Situated question answering in 3d scenes. In *ICLR*, 2023a.

683 Yecheng Jason Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy  
684 Zhang. Vip: Towards universal visual reward and representation via value-implicit pre-training. In  
685 *ICLR*, 2023b.

686 Lukas Mehl, Jenny Schmalfluss, Azin Jahedi, Yaroslava Nalivayko, and Andrés Bruhn. Spring: A  
687 high-resolution high-detail dataset and benchmark for scene flow, optical flow and stereo. In *CVPR*,  
688 2023.

689 Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3m: A universal  
690 visual representation for robot manipulation. In *CoRL*, 2022.

691 Yixin Nie, Xiang Zhou, and Mohit Bansal. What can we learn from collective human opinions on  
692 natural language inference data? In *EMNLP*, 2020.

693 Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov,  
694 Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning  
695 robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

696 Jishnu Jaykumar P, Yu-Wei Chao, and Yu Xiang. FewSOL: A Dataset for Few-Shot Object Learning  
697 in Robotic Environments. In *2023 IEEE International Conference on Robotics and Automation*  
698 (*ICRA*), pages 9140–9146, 2023. doi: 10.1109/ICRA48891.2023.10161143.

699 Jishnu Jaykumar P, Cole Salvato, Vinaya Bomnale, Jikai Wang, and Yu Xiang. iTeach: In the Wild  
700 Interactive Teaching for Failure-Driven Adaptation of Robot Perception, 2026. URL <https://arxiv.org/abs/2410.09072>.

702 Howard H. Qian, Yangxiao Lu, Kejia Ren, Gaotian Wang, Ninad Khargonkar, Yu Xiang, and Kaiyu  
703 Hang. Riseq: Robot interactive object segmentation via body frame-invariant features. In *ICRA*,  
704 2024.

705 Santhosh K Ramakrishnan, Aaron Gokaslan, Erik Wijmans, Oleksandr Maksymets, Alex Clegg, John  
706 Turner, Eric Undersander, Wojciech Galuba, Andrew Westbury, Angel X Chang, et al. HM3D:  
707 Habitat-matterport 3d dataset. In *NeurIPS Datasets and Benchmarks*, 2021.

708 Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham  
709 Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, Eric Mintun, Junting Pan, Kalyan Vasudev  
710 Alwala, Nicolas Carion, Chao-Yuan Wu, Ross Girshick, Piotr Dollár, and Christoph Feichtenhofer.  
711 Sam 2: Segment anything in images and videos, 2024. URL <https://arxiv.org/abs/2408.00714>.

713 Manolis Savva, Abhishek Kadian, Oleksandr Maksymets, Yili Zhao, Erik Wijmans, Bhavana Jain,  
714 Julian Straub, Jia Liu, Vladlen Koltun, Jitendra Malik, et al. Habitat: A platform for embodied AI  
715 research. In *ICCV*, 2019.

716 Pierre Sermanet, Tianli Ding, Jeffrey Zhao, et al. RoboVQA: Multimodal long-horizon reasoning for  
717 robotics. *arXiv preprint arXiv:2311.00899*, 2023.

718 Jiahao Shao, Yuanbo Yang, Hongyu Zhou, Youmin Zhang, Yujun Shen, Vitor Guizilini, Yue Wang,  
719 Matteo Poggi, and Yiyi Liao. Learning temporally consistent video depth from video diffusion  
720 priors. In *CVPR*, 2025.

721 Kaustubh Sridhar, Souradeep Dutta, Dinesh Jayaraman, and Insup Lee. Ricl: Adding in-context  
722 adaptability to pre-trained vision-language-action models. In *CoRL*, 2025.

723 Markus Suchi, Timothy Patten, David Fischinger, and Markus Vincze. Easylabel: A semi-automatic  
724 pixel-wise object annotation tool for creating robotic RGB-D datasets. In *ICRA*, 2019.

- 725 Pei Sun, Henrik Kretzschmar, Xerxes Dotiwalla, Aurelien Chouard, Vijaysai Patnaik, Paul Tsui, James  
726 Guo, Yin Zhou, Yuning Chai, Benjamin Caine, et al. Scalability in perception for autonomous  
727 driving: Waymo Open Dataset. In *CVPR*, 2020.
- 728 Rohan Taori, Achal Dave, Vaishal Shankar, Nicholas Carlini, Benjamin Recht, and Ludwig Schmidt.  
729 Measuring robustness to natural distribution shifts in image classification. In *NeurIPS*, 2020.
- 730 Chen Wang, Linxi Fan, Jiankai Sun, Ruohan Zhang, Li Fei-Fei, Danfei Xu, Yuke Zhu, and Anima  
731 Anandkumar. Mimicplay: Long-horizon imitation learning by watching human play. In *CoRL*,  
732 2023.
- 733 Tai Wang, Xiaohan Mao, Chenming Zhu, Runsen Xu, Ruiyuan Lyu, Peisen Li, Xiao Chen, Wenwei  
734 Zhang, Kai Chen, Tianfan Xue, Xihui Liu, Cewu Lu, Dahua Lin, and Jiangmiao Pang. Em-  
735 bodiedscan: A holistic multi-modal 3d perception suite towards embodied ai. In *CVPR*, pages  
736 19757–19767, 2024.
- 737 Alex Wong, Xiaohan Fei, Stephanie Tsuei, and Stefano Soatto. Unsupervised depth completion from  
738 visual inertial odometry. *IEEE Robotics and Automation Letters*, 5(2), 2020.
- 739 Yan Xia, Ran Ding, Ziyuan Qin, Guanqi Zhan, Kaichen Zhou, Long Yang, Hao Dong, and Daniel  
740 Cremers. TARGO: Benchmarking target-driven object grasping under occlusions. *arXiv preprint*  
741 *arXiv:2407.06168*, 2024.
- 742 Yu Xiang, Tanner Schmidt, Venkatraman Narayanan, and Dieter Fox. PoseCNN: A convolutional  
743 neural network for 6d object pose estimation in cluttered scenes. In *RSS*, 2018.
- 744 Rui Yang et al. Embodiedbench: Comprehensive benchmarking multi-modal large language models  
745 for vision-driven embodied agents. *arXiv preprint arXiv:2502.09560*, 2025.
- 746 Sriram Yenamandra, Arun Ramachandran, Karmesh Yadav, Austin Wang, Mukul Khanna, Theophile  
747 Gervet, Tsung-Yen Yang, Vidhi Jain, Alexander William Clegg, John Turner, et al. HomeRobot:  
748 Open-vocabulary mobile manipulation. In *CoRL*, 2023.
- 749 Yanjie Ze, Gu Zhang, Kangning Zhang, Chenyuan Hu, Muhan Wang, and Huazhe Xu. 3D diffusion  
750 policy: Generalizable visuomotor policy learning via simple 3D representations. In *Robotics:*  
751 *Science and Systems (RSS)*, 2024.
- 752 Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language  
753 image pre-training. *ICCV*, 2023.
- 754 Jiahui Zhang, Yusen Luo, Abrar Anwar, Sumedh Anand Sontakke, Joseph J Lim, Jesse Thomason,  
755 Erdem Biyik, and Jesse Zhang. Rewind: Language-guided rewards teach robot policies without  
756 new demonstrations. In *CoRL*, 2025.
- 757 Xinyu Zhang, Yuhao Liu, Yuting Wang, and Abdeslam Boularias. Detect everything with few  
758 examples. In *CoRL*, 2024.
- 759 Haoyu Zhen, Qiao Sun, Hongxin Zhang, Junyan Li, Siyuan Zhou, Yilun Du, and Chuang Gan.  
760 Tesseract: Learning 4d embodied world models. *ICCV*, 2025.
- 761 Ziyu Zhu, Xiaojian Ma, Yixin Chen, Zhidong Deng, Siyuan Huang, and Qing Li. 3D-VisTA:  
762 Pre-trained transformer for 3d vision and text alignment. In *ICCV*, 2023.

## 763 A Three-Phase Protocol: Extended Justification

764 **Clutter is the default.** A robot deployed in a home, warehouse, or lab almost always encounters  
765 cluttered scenes—this is the most frequent operating state [Liu et al., 2024a, Khazatsky et al., 2024].

**Interaction is the learning signal.** The fastest-growing paradigm in robot learning is training from human video—a trend spanning representations, policies, and world models. R3M [Nair et al., 2022] and VIP [Ma et al., 2023b] learn visual representations and reward functions from Ego4D [Grauman et al., 2022]; WHIRL [Bahl et al., 2022] learns one-shot imitation from unstructured human videos; MimicPlay [Wang et al., 2023] learns long-horizon manipulation by watching *human play*—closely mirroring R<sup>2</sup>PX’s interaction phase; Bharadhwaj et al. [Bharadhwaj et al., 2023] demonstrate zero-shot manipulation from passive human videos; Track2Act [Bharadhwaj et al., 2024] predicts point tracks from internet interaction videos; EgoMimic [Kareer et al., 2024] scales imitation via egocentric video; EgoDex [Hoque et al., 2025] trains dexterous policies from 829 hours of egocentric manipulation; DROID [Khazatsky et al., 2024] collects 76K in-the-wild episodes; and GR-2 [Cheang et al., 2024] trains VLAs directly from video. All process human interaction video as input—and perception robustness on this data strongly influences what the robot learns.

**Clean reveals recovery.** Robots rarely encounter tidy scenes, but the clean phase serves as a within-scene control and tests post-manipulation verification.

**Why exactly three.** Two phases would miss the critical interaction state. More than three adds diminishing returns: observe  $\rightarrow$  act  $\rightarrow$  verify covers the full manipulation cycle. Three phases yield three controlled comparisons ( $\mathcal{P}_{\text{clu}} \rightarrow \mathcal{P}_{\text{int}}, \mathcal{P}_{\text{int}} \rightarrow \mathcal{P}_{\text{cln}}, \mathcal{P}_{\text{clu}} \rightarrow \mathcal{P}_{\text{cln}}$ ) that feed the STR metric.

**What each phase reveals per task.** Table 7 shows the specific deployment condition each phase tests for each task.

Table 7: What each phase tests for each task.

|            | Clutter                               | Interaction                         | Clean                                      |
|------------|---------------------------------------|-------------------------------------|--------------------------------------------|
| D1 Depth   | Metric accuracy under occlusion       | Temporal flicker from hand entry    | Accuracy on simpler geometry (control)     |
| D2 Det.    | Detection under dense occlusion       | Detection during hand occlusion     | Reduced occlusion but changed spatial refs |
| D3 Track   | ID maintenance under camera motion    | ID survival through hand occlusion  | Re-identification after rearrangement      |
| D4 QA      | Counting & attribute QA under clutter | QA when hands occlude objects       | Reorganised scene (different answers)      |
| D5 Spatial | Spatial grounding in dense layout     | Spatial answers become <i>wrong</i> | New arrangement (tests adaptation)         |

## B ESD Feature Definitions

This appendix provides the full mathematical definitions of all 27 ESD features (7 categories) described in Section 3.2.

Given a scene  $s$  and phase  $p$ , let  $\mathcal{V}_{s,p} = \{I_t, D_t, M_t, \mathbf{T}_t, F_t^L, F_t^R\}_{t=1}^T$  denote the video sequence (RGB, D435 depth, instance mask, SE(3) pose, and T265 left/right fisheye images; the fisheye stream is optional—features default to zero when absent).

**Annotation effort.** Let  $\mathcal{K}_{s,p} = \{k_j\}$  be the mask refinement iteration counts:  $f_{\text{iter-mean}} = \frac{1}{|\mathcal{K}|} \sum_j k_j$ ,  $f_{\text{iter-max}} = \max_j k_j$ .

**Scene complexity.** Let  $\mathcal{O}_t$  be the visible objects in frame  $t$ ,  $N_s$  the total annotated objects:

$$f_{\text{obj-mean}} = \frac{1}{T} \sum_t |\mathcal{O}_t|, \quad f_{\text{obj-std}} = \sqrt{\frac{1}{T} \sum_t (|\mathcal{O}_t| - f_{\text{obj-mean}})^2} \quad (1)$$

$$f_{\text{obj-consist}} = \frac{1}{T} \sum_t \mathbb{1}[|\mathcal{O}_t| = N_s] \quad (2)$$

794 **Occlusion.**  $\text{occ}(o) = \sum_{o' \neq o} \text{area}(\text{bbox}(o) \cap \text{bbox}(o')) / \text{area}(\text{bbox}(o))$ .

$$f_{\text{occ-mean}} = \mathbb{E}_o[\text{occ}(o)], \quad f_{\text{occ-p90}} = P_{90}(\text{occ}(o)) \quad (3)$$

$$f_{\text{occ-heavy}} = \frac{1}{|\mathcal{O}|} \sum_o \mathbb{I}[\text{occ}(o) > 0.3] \quad (4)$$

**Depth quality.**

$$f_{\text{depth-invalid}} = \frac{1}{T} \sum_t \frac{\sum_x \mathbb{I}[D_t(x) \leq 0]}{|\Omega|} \quad (5)$$

$$f_{\text{depth-invalid-mask}} = \frac{1}{T} \sum_t \frac{\sum_{x \in M_t} \mathbb{I}[D_t(x) \leq 0]}{|M_t|} \quad (6)$$

$$f_{\text{depth-std}} = \frac{1}{T} \sum_t \text{Std}(D_t(x)), \quad x \in \text{valid} \quad (7)$$

$$f_{\text{depth-std-mask}} = \frac{1}{T} \sum_t \text{Std}(D_t(x)), \quad x \in M_t \cap \text{valid} \quad (8)$$

795 The whole-image  $f_{\text{depth-std}}$  is dominated by background variance;  $f_{\text{depth-std-mask}}$  isolates the object-  
796 relevant noise that grasp planning consumes.

797 **Photometric–depth conflict.**  $f_{\text{specular}} = \frac{1}{T} \sum_t \frac{\sum_x \mathbb{I}[I_t(x) > 230 \wedge D_t(x) \text{ invalid}]}{|\Omega|}$ ;  $f_{\text{dark-conflict}} =$   
798  $\frac{1}{T} \sum_t \frac{\sum_x \mathbb{I}[I_t(x) < 30 \wedge D_t(x) \text{ invalid}]}{|\Omega|}$ . The two features capture the dominant D435 failure modes (trans-  
799 parent/reflective surfaces and out-of-range / poorly-lit regions) — pixels that look fine in RGB but  
800 where the depth sensor reports zero. The conflict suffix disambiguates from  $f_{\text{fisheye-dark}}$  below, which  
801 is a pure-intensity statistic on the T265 stream.

802 **Temporal annotation stability.** Let  $A_t^o$  be the per-frame pixel area of object  $o$  and  $v_t^o \in \{0, 1\}$  its  
803 visibility:

$$\text{CV}(o) = \sigma(A^o) / \mu(A^o); \quad f_{\text{area-cv}} = \mathbb{E}_o[\text{CV}(o)] \quad (9)$$

$$f_{\text{area-drop}} = \mathbb{E}_o \left[ \frac{1}{T} \sum_{t=2}^T \mathbb{I} \left[ \frac{A_{t-1}^o - A_t^o}{A_{t-1}^o} > \delta \right] \right], \quad \delta = 0.5 \quad (10)$$

$$f_{\text{vis-instability}} = \mathbb{E}_o \left[ \frac{1}{T} \sum_{t=2}^T \mathbb{I}[v_t^o \neq v_{t-1}^o] \right] \quad (11)$$

804  $f_{\text{area-drop}}$  counts sudden mask collapses ( $>50\%$  area loss between consecutive frames), typically  
805 caused by an arm sweeping across the object during the interaction phase.

806 **Camera motion.** Let  $\Delta \mathbf{t}_t = \|\mathbf{t}_{t+1} - \mathbf{t}_t\|$  and  $\Delta \mathbf{R}_t = \text{angle}(\mathbf{R}_t^{-1} \mathbf{R}_{t+1})$ :

$$f_{\text{trans-mean}} = \mathbb{E}[\Delta \mathbf{t}_t], \quad f_{\text{trans-p90}} = P_{90}(\Delta \mathbf{t}_t) \quad (12)$$

$$f_{\text{rot-mean}} = \mathbb{E}[\Delta \mathbf{R}_t], \quad f_{\text{rot-p90}} = P_{90}(\Delta \mathbf{R}_t) \quad (13)$$

$$f_{\text{jerk}} = \mathbb{E}[|\Delta \mathbf{t}_{t+1} - \Delta \mathbf{t}_t|] \quad (14)$$

807 The  $P_{90}$  features capture heavy-tail motion spikes; the means alone hide brief but extreme camera  
808 jerks during interaction.

809 **Fisheye / stereo features.** The T265 ships a synchronised left/right fisheye pair  $(F_t^L, F_t^R)$ . We  
810 compute five statistics that are independent of D435 calibration; the features default to zero when the  
811 fisheye stream is absent. Let  $\tau_b = 230$ ,  $\tau_d = 30$  match the photometric thresholds above.

$$f_{\text{fisheye-dark}} = \mathbb{E}_t \left[ \frac{\sum_x \mathbb{I}[F_t^L(x) < \tau_d]}{|\Omega|} \right], \quad f_{\text{fisheye-bright}} = \mathbb{E}_t \left[ \frac{\sum_x \mathbb{I}[F_t^L(x) > \tau_b]}{|\Omega|} \right] \quad (15)$$

$$f_{\text{fisheye-sharpness}} = \mathbb{E}_t [\text{Var}(\nabla^2 F_t^L)] \quad (16)$$

$$f_{\text{fisheye-corr}} = \mathbb{E}_t [\text{corr}(F_t^L, F_t^R)] \quad (17)$$

$$f_{\text{fisheye-texture}} = \mathbb{E}_t [\|\nabla F_t^L\|] \quad (18)$$

812  $f_{\text{fisheye-sharpness}}$  is the variance-of-Laplacian blur measure; low values indicate motion blur.  $f_{\text{fisheye-corr}}$   
813 is the Pearson correlation between left and right images (pre-rectification, resized to common  
814 shape if needed); transparent or reflective surfaces, and view-dependent specularities, drive it down.  
815  $f_{\text{fisheye-texture}}$  is the mean per-pixel gradient magnitude — proxy for textural information available to  
816 feature-matching pipelines.

817 **Raw feature data.** The full  $297 \times 27$  matrix of percentile-normalised feature values is shipped in  
818 the splits manifest (benchmark/data/splits/phase\_esd\_splits.csv) and visualised in three  
819 views: Figure 7 hierarchically clusters every scene against every feature and reveals the natural  
820 difficulty groupings the methodology recovers; Figure 8 shows the per-modality fingerprint of each  
821 Easy / Medium / Hard tier, making visible exactly which categories push a scene into each tier;  
822 Figure 9 collapses the 27-dimensional matrix onto the single 1-D RPX-DS score axis on which the  
823 tertile cuts are actually made, exposing the cut surface itself.

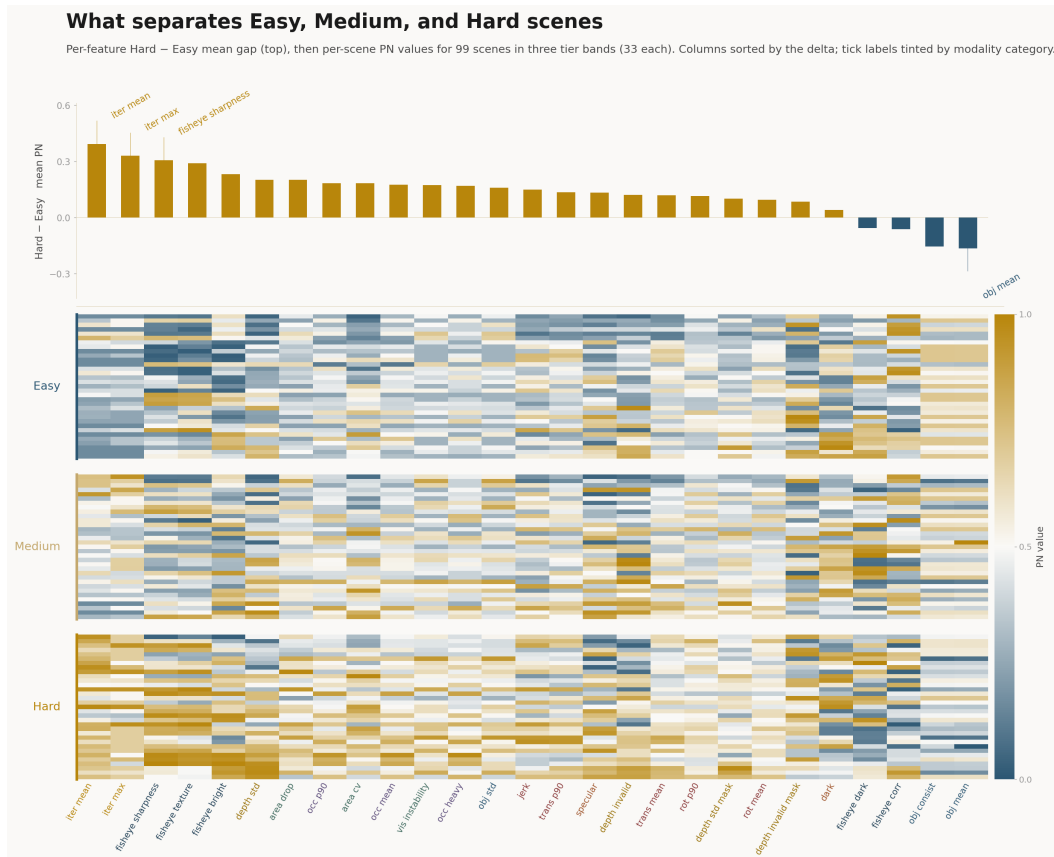


Figure 7: **What separates Easy, Medium, and Hard scenes.** *Top:* per-feature HARD – EASY mean PN gap, sorted descending. Red bars (positive gap) mark features that are systematically elevated in Hard scenes; teal bars (negative gap) mark features that lean toward Easy. The three most Hard-discriminative features are annotated inline. *Bottom:* per-scene PN values for all 99 scenes, split into three horizontal bands by primary effort\_stratified\_v1 tier (33 scenes each, by tertile cut on the per-scene combined RPX-DS score). Columns are in the same order as the delta bar above; the diverging colormap is centred on the dataset median (0.5), so above-median cells render warm and below-median cells render cool. Visual reading: the Hard band runs warm on the left (where the delta is large and positive) and the Easy band runs cool there, with both bands converging toward the right where the delta crosses zero. X-axis tick labels are tinted by modality category.

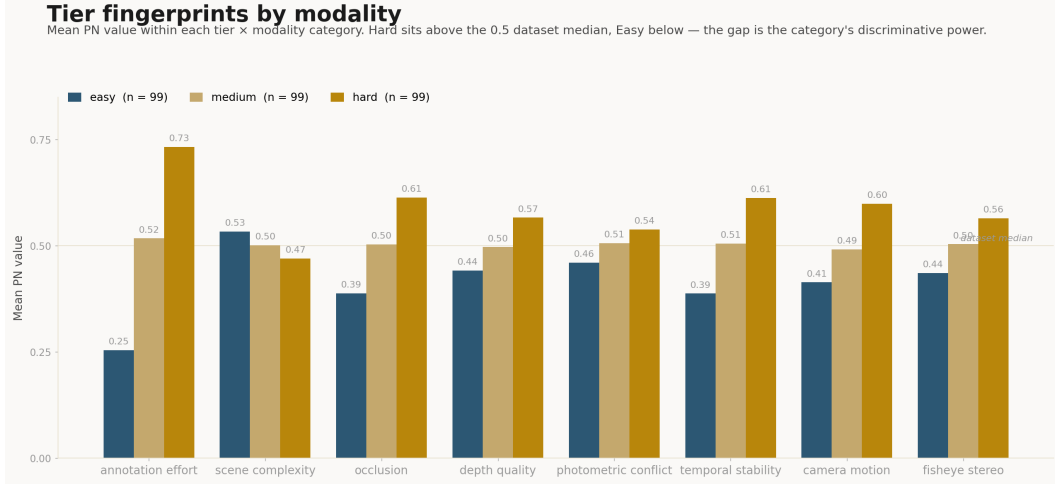


Figure 8: **Per-tier fingerprint — which features pull a scene into each tier.** Mean percentile-normalised feature value within each tier, with the dataset median (0.5) marked. Easy entries (top) sit predominantly below 0.5 across most features; Hard entries (bottom) sit predominantly above 0.5. Annotation-effort features (leftmost two bars, blue) show the largest tier-to-tier swing, which is consistent with the effort-anchored weighting; perception features show the next-largest swings, with depth quality and occlusion notably elevated in the Hard tier.

## 824 C ESD Weight Sensitivity Analysis

825 We sample 1,000 weight vectors  $w$  uniformly from the simplex (Dirichlet(1)) over all 27 fea-  
 826 tures and recompute the RPX-DS ranking under each perturbed weight vector. Compared to the  
 827 uniform RPX-DS, the median Kendall  $\tau$  between the perturbed and uniform rankings is 0.685  
 828 (IQR: 0.630–0.734), demonstrating that the relative difficulty ordering is largely preserved across  
 829 scoring choices. At the tier-assignment level, 88.6% of (scene, phase) pairs retain their domi-  
 830 nant tier (Easy/Medium/Hard) across the 1,000 perturbations; 11.4% of pairs flip tier in fewer  
 831 than 5% of perturbations (essentially fully stable), and only 33 of 297 pairs (11.1%) flip in  
 832 more than half of perturbations. The full per-pair stability table is shipped in the splits manifest  
 833 (benchmark/data/splits/methodology\_study/stability/weight\_perturbation.csv).

## 834 D Splits Methodology Study

835 To validate the choice of uniform RPX-DS as the primary scoring method, we con-  
 836 ducted a seven-layer methodology study comparing 11 scoring methods spanning ev-  
 837 ery major family on the released 297 (scene, phase) feature table. The study is  
 838 fully reproducible from the dataset alone (no model results required) and is shipped as  
 839 experiments/neurips-2026/scripts/difficulty\_methodology\_study.py.

840 **Methods compared.** *Linear aggregation on percentile-normalised features:* mean (uniform\_v1,  
 841 our primary), median, max, top-3 mean. *Dimensionality reduction:* first principal component (PCA-  
 842 PC1). *Clustering (mapped to 3 tiers via centroid magnitude):* k-means at  $k \in \{3, 5, 8\}$ , Gaussian  
 843 mixture ( $k = 3$ , full covariance), agglomerative Ward linkage ( $k = 3$ ), and DBSCAN with  $\epsilon$  chosen  
 844 via the  $k$ -distance knee. For visualisation, we additionally compute t-SNE, Locality Preserving  
 845 Projections (LPP, He and Niyogi [2003]), and Spectral Embedding 2D projections.

846 **L1: Cross-method agreement.** Median pairwise Kendall  $\tau$  between continuous scores: 0.363. Me-  
 847 dian pairwise Adjusted Rand Index (ARI) between categorical tiers: 0.083. Cross-family agreement  
 848 is generally low—e.g.,  $\tau(\text{mean\_pn}, \text{max\_pn}) = 0.24$ ,  $\tau(\text{mean\_pn}, \text{PCA-PC1}) = 0.39$ —confirming  
 849 that scoring choice is methodologically consequential and motivating the consensus-tier reporting in  
 850 §3.2.

## The 1-D axis on which Easy/Medium/Hard cuts are made

Effort-stratified convex score ( $\alpha = 0.25$ ) over 27 percentile-normalised features. Dashed lines mark the 33% / 67% percentile cuts; shading marks the tier region.

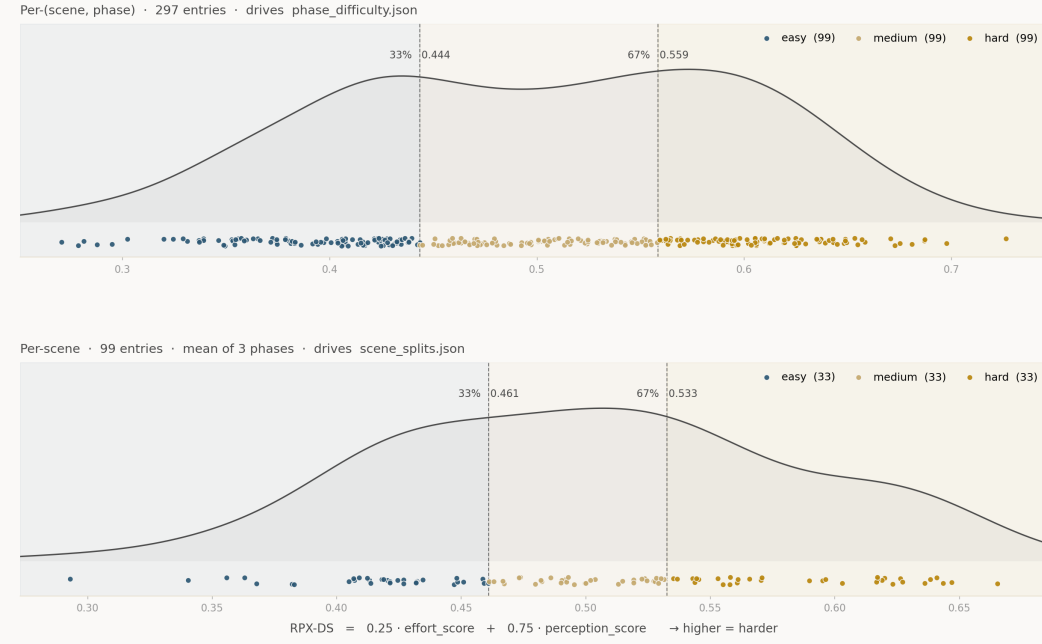


Figure 9: **The 1-D axis on which Easy/Medium/Hard cuts are made.** The 27-feature percentile-normalised matrix is collapsed to a single scalar via the effort-stratified convex score  $\text{RPX-DS}(s, p) = 0.25 \cdot \text{effort\_score} + 0.75 \cdot \text{perception\_score}$ , and the 33% / 67% percentile cut points (dashed lines) define the Easy / Medium / Hard boundaries. *Top*: per-(scene, phase) view — 297 entries, the level features are extracted at; tiers here drive `phase_difficulty.json`. *Bottom*: per-scene view — 99 entries, computed as the mean of the 3 per-phase scores per scene; tiers here drive `scene_splits.json`. Coloured dots show the assigned tier of each entry; band shading marks the tier region. The per-scene distribution is necessarily tighter than the per-phase one (averaging 3 phases shrinks variance), but the overall unimodal shape and the fact that the cuts sit on relatively gentle slopes — rather than at sharp valleys — is what motivates the stability tests in §C: small score perturbations near the cuts can flip a few entries’ tier assignments without invalidating the global ordering.

851 **L2: Number of clusters.** K-means silhouette is maximised at  $k = 2$  (silhouette = 0.293); GMM  
852 silhouette is maximised at  $k = 3$  (0.238). Tibshirani’s gap statistic also recommends  $k = 2$  (Fig. 10).  
853 The data therefore does *not* natively support a 3-cluster structure; the 33/33/34 tertile is adopted as a  
854 reporting convention because Easy/Medium/Hard is canonical in benchmarks.

855 **L3: Stability under perturbations.** Three independent perturbation studies on `uniform_v1` (1,000  
856 iterations each unless noted): (a) *Bootstrap subsampling (80%)*: median per-pair dominant-tier  
857 stability = 1.000; 89.9% of pairs have >95% stability across resamples; 81.1% of pairs are perfectly  
858 stable. (b) *Weight perturbation (Dirichlet(1))*: median tier-flip rate 0.348; 88.6% of pairs retain  
859 dominant tier (see §C for the full Kendall- $\tau$  analysis). (c) *Feature dropout (drop 1/3 random features,*  
860 *200 iterations)*: median tier-flip rate 0.275; 23.9% of pairs flip in <5% of dropouts. The full per-pair  
861 distributions are in Fig. 11.

862 **L4: Outliers.** Twelve (scene, phase) pairs exceed the  $\chi^2_{27}(0.99)$  Mahalanobis-distance threshold in  
863 the percentile-normalised feature space. The top entries by distance are `scene100.jsom.atrium`  
864 `interaction phase` ( $d = 11.95$ ), `scene18.su.CHECKERBOARD clutter phase` ( $d = 10.86$ ), and  
865 `scene74.ecsw.atriumStairs clean phase` ( $d = 7.69$ ). Two of these (`scene100` and `scene74`)  
866 coincide with documented T265 tracking-failure events during capture, demonstrating that the  
867 methodology surfaces genuine data quality issues from the feature distribution alone.

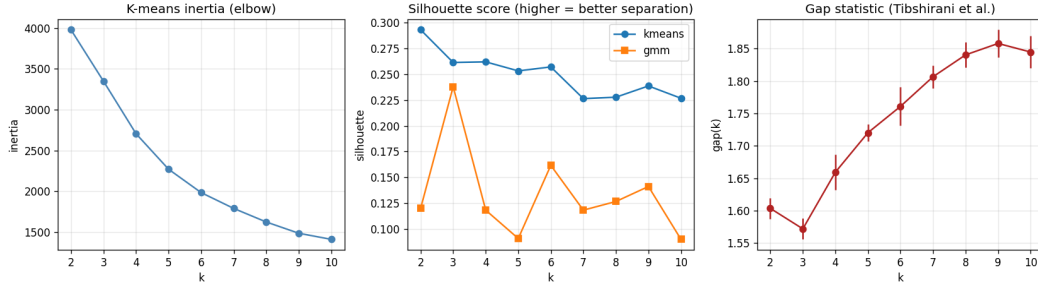


Figure 10: **Cluster-count diagnostics on the 297-pair, 27-feature matrix.** K-means inertia (left) shows no sharp elbow. K-means silhouette (centre, blue) peaks at  $k = 2$ ; GMM silhouette (centre, orange) peaks at  $k = 3$ . Tibshirani’s gap statistic (right) is monotonically increasing through  $k = 10$ ; the smallest  $k$  satisfying  $gap(k) \geq gap(k + 1) - s_{k+1}$  is  $k = 2$ . The data does not natively prefer  $k = 3$  — the tertile is a reporting convention.

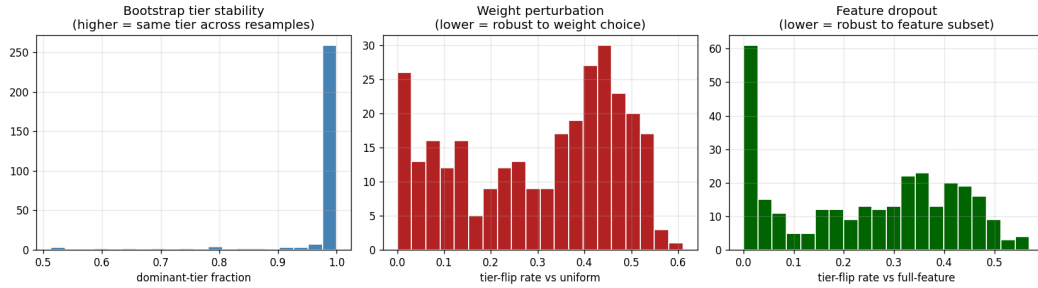


Figure 11: **Stability of the uniform\_v1 tier assignment under three perturbation studies.** Bootstrap subsampling (left, blue) is dominated by the mass at 1.0 — the assignment is essentially invariant to which 80% of scenes is sampled. Weight perturbation (centre, red) is bimodal: a stable subpopulation flips little and a sensitive subpopulation flips  $\sim 50\%$  of the time. Feature dropout (right, green) shows a stable mode at 0 and a long tail. Bootstrap-rock-stable but weight-sensitive — the case for shipping the consensus tier alongside any single primary method.

868 **L5: Cross-method consensus.** Across the 11 methods (DBSCAN excluded as degenerate—it  
869 found no density modes in the 27D feature space), only 24 of 297 pairs (8.1%) receive unanimous  
870 tier assignment. 104 pairs (35.0%) are split between two tiers, and 169 pairs (56.9%) span all three  
871 (Fig. 12). The consensus tier (majority vote) is shipped alongside uniform\_v1 in the splits manifest.

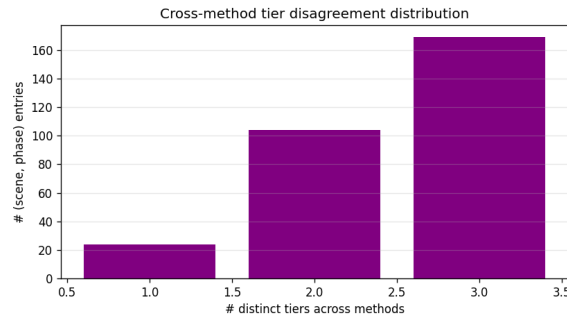


Figure 12: **Cross-method tier disagreement.** Histogram of distinct tiers assigned per (scene, phase) across 11 scoring methods. Only 8.1% of pairs receive unanimous agreement; the majority span 2 or 3 tiers. The consensus tier (majority vote) is shipped alongside the primary uniform\_v1 tier.

872 **L6: Phase signature.** The maximum ARI between any clustering method’s labels and the capture-  
873 phase index (clutter/interaction/clean) is 0.182 (k-means at  $k = 8$ ); for  $k = 3$  partitions specifically: k-

means 0.177, GMM 0.036, Ward 0.003, DBSCAN 0.003. Feature space therefore captures perception difficulty rather than capture-protocol identity—validating that ESD splits are a difficulty stratification, not a phase classifier.

**Per-feature contribution by tier.** Fig. 13 shows the mean percentile-normalised value of each feature within each Easy/Medium/Hard tier. Hard scenes are higher than Easy on most features—confirming the score is broadly difficulty-tracking rather than driven by a single modality. Three features show inverted ordering:  $f_{\text{obj-mean}}$  and  $f_{\text{obj-consist}}$  (densely-arranged static scenes are “Easy” in the multi-feature sense), and  $f_{\text{fisheye-corr}}$  (transparent or motion-blurred scenes break stereo correlation, marking them as Hard with low corr).

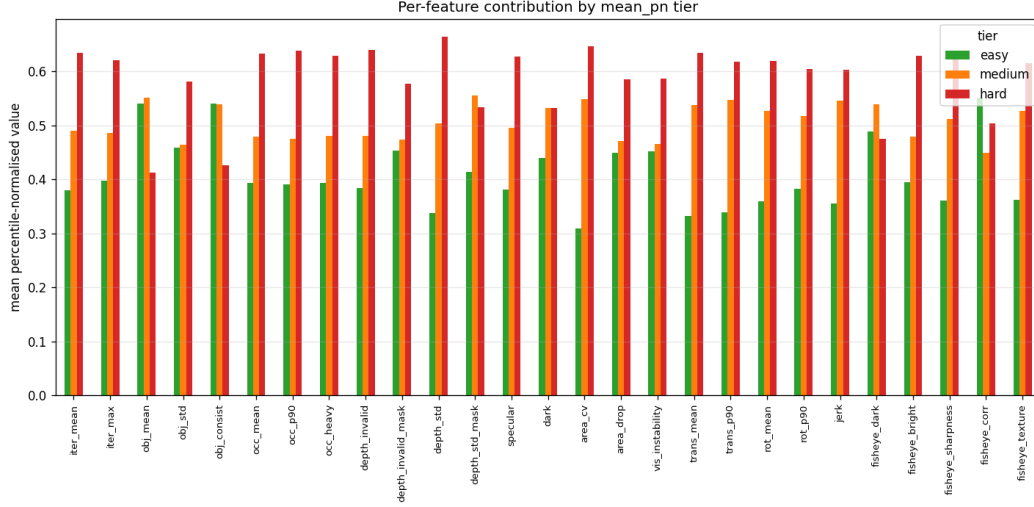


Figure 13: **Per-feature mean percentile-normalised value by uniform\_v1 tier.** Hard tier (red) is uniformly higher than Easy (green) on most features, confirming the multi-modal nature of the difficulty signal. Three inversions ( $f_{\text{obj-mean}}$ ,  $f_{\text{obj-consist}}$ ,  $f_{\text{fisheye-corr}}$ ) reveal that scene complexity  $\neq$  scene difficulty: densely arranged but well-lit static scenes can still be Easy, while sparsely arranged but transparent or moving scenes are Hard.

**Reproducibility.** All numbers above are deterministic given seed=0; the full study (tables, plots, and per-entry assignments under all 11 methods) ships under benchmark/data/splits/methodology\_study/, with input file SHA-256 stamped into provenance.json.

## E Ground-Truth Annotation Pipeline

This section documents the three major annotation pipelines in detail. Figure 14 illustrates the mask annotation workflow, Figure 15 illustrates the QA ground-truth generation, and Figure 16 illustrates the camera pose refinement procedure.

### E.1 Instance Mask Annotation

**TODO:** Figure: mask annotation pipeline diagram — SAM2 auto → GroundingDINO bbox proposals → human iterative refinement UI → final masks. Show example frames at each stage.

Figure 14: Instance mask annotation pipeline. (1) SAM2 auto-mode generates initial masks. (2) GroundingDINO proposes bounding boxes for missed objects using text prompts from the object questionnaire. (3) A human annotator iteratively refines boundaries in a custom UI until sub-pixel accuracy at object edges. The refinement iteration count  $k_j$  per instance feeds ESD (§3.2).

SAM2 [Ravi et al., 2024] in automatic mode generates initial instance masks for each keyframe. GroundingDINO [Liu et al., 2024b] provides complementary bounding-box proposals using per-object category names from the FewSQL questionnaire, catching objects that SAM2’s class-agnostic mode misses (e.g. transparent items, thin utensils). A custom annotation interface then enables iterative human refinement: annotators adjust mask boundaries, split merged instances, and add missing objects until edges align with object contours at sub-pixel accuracy. The refinement iteration count  $k_j$  per instance is recorded and directly feeds the ESD difficulty model (Section 3.2)—scenes requiring more human correction are empirically harder for models.

**Tracklets.** SAM2 video propagation with human-corrected keyframe initialisation produces dense tracklets with consistent IDs across each phase. Annotators verify ID consistency at phase boundaries and correct drift or ID switches introduced by fast hand motion during interaction.

## E.2 QA Ground-Truth Generation

**TODO:** Figure: QA generation pipeline — (a) human-authored template bank with robotics-grounded question types, (b) per-frame instantiation using visibility masks + bbox geometry + questionnaire attributes, (c) example Q/A pairs showing phase-varying answers.

Figure 15: QA ground-truth generation. Humans author question templates grounded in robotics operator needs (left). Templates are instantiated deterministically per frame using object visibility masks, bounding-box geometry, and questionnaire attributes (centre). The same question produces different correct answers across phases (right).

**Template design.** A set of question templates was authored by hand to reflect the information needs of a robot operator or planner. Templates fall into four categories: **(i) Identity:** “What objects are on the table?”, “Is there a {category}?”; **(ii) Attribute verification:** “What colour is the {object}?”, “What material is the {object}?”; **(iii) Counting:** “How many objects are visible?”; **(iv) Spatial relations (D5):** “What is to the left/right/above/below of the {object}?”. All templates use crowd-sourced natural-language object descriptions from FewSQL questionnaires [P et al., 2023] (§4), matching how real users command robots.

**Instantiation.** For each frame  $t$ , templates are instantiated deterministically: (a) per-frame visibility is determined from instance masks (an object is “visible” if its mask area exceeds a minimum threshold); (b) identity and attribute answers are read from the questionnaire; (c) spatial relations are computed in the *image plane* using polar coordinates centred on the reference object’s bounding-box centroid. This yields **TODO: X** Q/A pairs per scene across all phases. Critically, the same question has *different correct answers* across phases (e.g. objects are rearranged, new objects become visible), making this, to our knowledge, the first perception QA benchmark with phase-varying ground truth.

**Spatial relation definition.** All spatial relations are defined in the image plane: “left” = centroid with smaller  $x$ -coordinate; “above” = smaller  $y$ -coordinate; “near” = Euclidean distance  $< \tau_{\text{near}}$  pixels. This matches how VLMs process images and avoids ambiguity from world-frame perspective.

## E.3 Camera Pose Refinement

**TODO:** Figure: pose refinement pipeline — T265 raw VIO trajectory → outlier rejection → optional loop closure → refined trajectory. Show before/after trajectory overlay on a sample scene.

Figure 16: Camera pose refinement. Raw T265 VIO at 200 Hz is downsampled to 30 fps, filtered for outlier jumps, and validated via loop-closure analysis on closed-loop sequences.

T265 VIO provides 6-DoF camera pose at 200 Hz, downsampled to 30 fps to align with D435 frames. Raw T265 odometry is used as the pose reference—no offline bundle adjustment is applied, preserving the noise profile that deployed robots would encounter with the same sensor. Over  $\sim 250$  frames ( $\sim 8$  s per phase), drift is bounded to  $< \text{TODO: X cm} / < \text{TODO: X}^\circ$  via loop-closure analysis on **TODO: N** closed-loop sequences where the camera returns to its starting position. Outlier frames with  $\Delta t > \text{TODO: X cm}$  or  $\Delta R > \text{TODO: X}^\circ$  between consecutive frames are flagged and excluded from temporal stability (TS) computation.

929 **Depth ground truth.** Raw D435 structured-light depth; invalid pixels (specular reflection, trans-  
930 parency, range limits) are flagged and excluded from evaluation. No inpainting is applied—models  
931 are evaluated only on pixels where the sensor produces valid returns.

932 **Language annotations.** Per-object attributes (category, colour, material, shape) from FewSOL [P  
933 et al., 2023] questionnaires serve as referring expressions for grounding evaluation and as the  
934 vocabulary source for QA template instantiation (§E.2).

## 935 **F Per-Task Detailed Results**

936 **TODO:** One table per task (D1–D5) with: All models (including appendix-only: DA-V2 Base,  
937 Prompt DA, MiDaS, SAM2 Small, DITO, RO-ViT, etc.). Per-ESD columns (Easy / Medium / Hard).  
938 Per-phase columns ( $\mathcal{P}_{\text{clu}}$  /  $\mathcal{P}_{\text{int}}$  /  $\mathcal{P}_{\text{cIn}}$ ). Full metric suite (not just primary metric). Params, FLOPs,  
939 median latency.

## 940 **G Camera Motion Statistics**

941 **TODO:** Full-page figure: (a) Histogram of  $\Delta t$  (cm/frame) across 100 scenes  $\times$  3 phases. (b)  
942 Histogram of  $\Delta R$  (deg/frame). (c) Histogram of jerk. (d) Overlay with published platform motion  
943 profiles from Stretch RE2, Unitree H1, Fetch, Spot.

## 944 **H Dataset Object Taxonomy**

945 **TODO:** Full object taxonomy:  $\sim 70$  categories with supercategory assignments, instance counts,  
946 YCB/FewSOL overlap markers. Example images.

## 947 **I Text vs. Visual Prompting: Extended Analysis**

948 **TODO:** Extended comparison of text-prompted (GroundingDINO, OWL-ViT, YOLO-World, Qwen2-  
949 VL) vs. image-prompted (DE-ViT, NIDS-Net, Qwen2-VL with visual prompt) detection. Per-ESD,  
950 per-phase breakdown. Analysis of which object categories benefit most from each prompting mode.  
951 Failure case examples: text prompt fails when spatial relations change; visual prompt fails under  
952 viewpoint shift.

## 953 **J RPX Toolkit: Extended Documentation**

954 This appendix provides extended documentation for the rpx-benchmark toolkit described in Sec-  
955 tion 5.

### 956 **Python API usage example.**

```
957 from rpx_benchmark import (  
958     RPXDataset, BenchmarkRunner,  
959     MetricSuite, TaskType  
960 )  
961  
962 dataset = RPXDataset.from_manifest("rpx_test.json")  
963 model = MyDepthModel() # subclasses BenchmarkModel  
964 suite = MetricSuite.for_task(TaskType.MONOCULAR_DEPTH)  
965 runner = BenchmarkRunner(model, dataset, suite)  
966  
967 result, dr = runner.run_with_deployment_readiness(  
968     primary_metric="abs_rel",  
969     model_name="MyDepthModel"  
970 )  
971 print(dr.summary())
```

972 The dr.summary() output:

```
973 --- RPX Deployment Readiness: MyDepthModel ---
974 S_overall = 0.134 (AbsRel, lower=better)
975 STR_{C->I} = -0.023 (interaction drop)
976 STR_{I->L} = +0.018 (recovery)
977 TS = 0.891 (temporal stability)
978 SGC = 0.672 (geometric coherence)
979 Params = 335M FLOPs = 211G
980 Latency = 45ms (median, per frame)
```

981 **Adding a new task.**

```
982 from rpx_benchmark.tasks import register_task, TaskSpec
983 register_task(TaskSpec(
984     name="affordance_prediction",
985     runner=my_affordance_runner,
986     metrics=["auc", "f1"],
987     input_modalities=["rgb", "depth"],
988 ))
```

989 **Adding a new metric.**

```
990 from rpx_benchmark.metrics import BaseMetric, register_metric
991 class GraspAlignment(BaseMetric):
992     def compute(self, pred_mask, gt_depth):
993         return score
994 register_metric("grasp_alignment", GraspAlignment)
```

995 **CI.** 138 offline tests, Python 3.10/3.11/3.12, GitHub Actions, zero network deps. MkDocs +  
996 mkdocstrings auto-docs. Ruff lint.

## 997 **K Cross-Task Ranking Agreement**

998 **TODO:** (a) Kendall  $\tau$  of  $S_{\text{overall}}$  rankings for task pairs sharing  $\geq 2$  models. (b) Kendall  $\tau$  of scene  
999 difficulty rankings across tasks (ESD consistency). (c) For shared-backbone models: does deployment  
1000 readiness transfer across tasks?

## 1001 **L Croissant Metadata and Datasheet**

1002 **RPX** is documented using Croissant [Akhtar et al., 2024] with core and RAI fields, as required by  
1003 NeurIPS 2026 E&D Track. **Core:** name, URL, licence (CC BY 4.0), format, record sets for each  
1004 modality. **RAI:** collection protocol, consent, PII assessment (none), annotation methodology, known  
1005 biases (indoor-dominant, D435 range, human motion profile). The Croissant JSON is included with  
1006 the supplementary materials and hosted on HuggingFace.

## 1007 **M Scope, Assumptions, and Output-vs-Feature Evaluation**

1008 **RPX** is designed to test the claim: *pretrained perception backbones differ in deployment robustness,*  
1009 *and this difference is not predictable from standard accuracy alone.*

1010 **Output-level vs. feature-level evaluation.** Robot learning systems consume perception at two  
1011 levels. **(a) As raw signals:** modular systems (OK-Robot, DP3, AnyGrasp [Liu et al., 2024a, Ze  
1012 et al., 2024]) directly consume depth maps, bounding boxes, and masks—for these, **RPX** evaluates  
1013 the signal the robot directly consumes. **(b) As encoded features:** VLAs ( $\pi_0$ , OpenVLA, GR00T)  
1014 consume frozen or fine-tuned encoder activations, never producing an explicit depth map or detection.  
1015 For these, **RPX**’s task-level metrics serve as *proxies* for feature quality: a backbone whose outputs

1016 exhibit low temporal stability (TS) or poor phase-transition robustness (STR) has representations  
1017 that fail to stably encode scene geometry or handle occlusion, and an action-conditioned decoder  
1018 is unlikely to fully recover information the encoder failed to capture. Furthermore, R<sup>P</sup>X’s VLM  
1019 evaluation (D4/D5) directly tests the same VLMs (PaliGemma, Qwen2-VL) that VLAs are built on.  
1020 The downstream validation in §6.7 empirically tests whether R<sup>P</sup>X’s rankings transfer to task-level  
1021 success.

1022 **Additional assumptions.** (i) Human-carried capture is a valid motion proxy (validated in §4  
1023 and Appendix G). (ii) 100 scenes are representative of deployment environments. (iii) Zero-shot  
1024 evaluation reflects deployment conditions.

1025 **Out of scope.** Navigation-specific perception, sensing outside the D435/T265 envelope (0.3–3 m  
1026 indoor + matched outdoor), and platforms with non-humanoid kinematics (e.g., fixed-base arms with  
1027 restricted viewpoints).

## 1028 NeurIPS Paper Checklist

### 1029 1. Claims

1030 Question: Do the main claims made in the abstract and introduction accurately reflect the  
1031 paper’s contributions and scope?

1032 Answer: [Yes]

1033 Justification: The abstract and introduction state that **RPX** is a unified RGB-D bench-  
1034 mark with five perception tasks, three-phase capture, and ESD-stratified difficulty splits.  
1035 Sections 4–6 directly support each claim with dataset statistics, metric definitions, and  
1036 experimental results. Scope limitations (indoor-only, D435 depth range, human-driven  
1037 capture trajectories) are stated in Section 4 and the conclusion.

### 1038 2. Limitations

1039 Question: Does the paper discuss the limitations of the work performed by the authors?

1040 Answer: [Yes]

1041 Justification: The conclusion contains a dedicated *Limitations* paragraph discussing T265  
1042 odometry drift, D435 depth range constraints, indoor-only capture, and annotator diversity.  
1043 The dataset card (Croissant) repeats these.

### 1044 3. Theory assumptions and proofs

1045 Question: For each theoretical result, does the paper provide the full set of assumptions and  
1046 a complete (and correct) proof?

1047 Answer: [N/A]

1048 Justification: The paper presents an empirical benchmark and dataset; it contains no theoret-  
1049 ical results or proofs.

### 1050 4. Experimental result reproducibility

1051 Question: Does the paper fully disclose all the information needed to reproduce the main ex-  
1052 perimental results of the paper to the extent that it affects the main claims and/or conclusions  
1053 of the paper (regardless of whether the code and data are provided or not)?

1054 Answer: [Yes]

1055 Justification: The evaluation toolkit is pip-installable (`rpX-benchmark`), open-source (MIT),  
1056 and includes a continuous-integration test suite verifying reproducibility of all metric  
1057 computations. Sensor specifications, annotation protocol details, and evaluation setup are  
1058 described in Sections 4 and 5. All model checkpoints are public.

### 1059 5. Open access to data and code

1060 Question: Does the paper provide open access to the data and code, with sufficient instruc-  
1061 tions to faithfully reproduce the main experimental results, as described in supplemental  
1062 material?

1063 Answer: [Yes]

1064 Justification: The dataset is hosted on HuggingFace Datasets under CC BY 4.0 with a  
1065 5-scene reviewer sample available without registration. The evaluation toolkit source code is  
1066 publicly available under the MIT licence with pip-installable dependencies and a documented  
1067 CLI.

### 1068 6. Experimental setting/details

1069 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-  
1070 rameters, how they were chosen, type of optimizer) necessary to understand the results?

1071 Answer: [Yes]

1072 Justification: Section 4 describes the train/val/test splits (70/15/15 scenes, no overlap),  
1073 Section 3.2 defines ESD computation and thresholds, and Section 6 reports GPU type,  
1074 inference settings, and per-model checkpoint identifiers. No training or fine-tuning is  
1075 performed; all models are evaluated zero-shot.

### 1076 7. Experiment statistical significance

1077 Question: Does the paper report error bars suitably and correctly defined or other appropriate  
1078 information about the statistical significance of the experiments?

1079 Answer: [Yes]

1080 Justification: Tables report standard deviation across scenes alongside mean values. The  
1081 ESD sensitivity analysis (Appendix A) uses 1,000 Monte Carlo weight perturbations to  
1082 assess split stability.

1083 **8. Experiments compute resources**

1084 Question: For each experiment, does the paper provide sufficient information on the com-  
1085 puter resources (type of compute workers, memory, time of execution) needed to reproduce  
1086 the experiments?

1087 Answer: [Yes]

1088 Justification: Section 6 states GPU type, VRAM, and per-model inference time. The toolkit  
1089 automatically reports FLOPs and median per-sample latency in every run.

1090 **9. Code of ethics**

1091 Question: Does the research conducted in the paper conform, in every respect, with the  
1092 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

1093 Answer: [Yes]

1094 Justification: The dataset contains indoor daily objects; no faces, biometric identifiers, or  
1095 personally identifiable information are captured. All capture was performed by lab personnel  
1096 who provided informed consent for data release. The work does not raise dual-use concerns.

1097 **10. Broader impacts**

1098 Question: Does the paper discuss both potential positive societal impacts and negative  
1099 societal impacts of the work performed?

1100 Answer: [Yes]

1101 Justification: The conclusion includes a *Broader Impact* paragraph discussing both the  
1102 positive impact (enabling safer robot deployment through rigorous perception evaluation)  
1103 and the limited risk profile (indoor daily-object scenes with no misuse pathway).

1104 **11. Safeguards**

1105 Question: Does the paper describe safeguards that have been put in place for responsible  
1106 release of data or models that have a high risk for misuse (e.g., pre-trained language models,  
1107 image generators, or scraped datasets)?

1108 Answer: [N/A]

1109 Justification: The dataset contains only indoor tabletop scenes with daily objects. It does not  
1110 contain faces, personal data, generative models, or any content that poses a misuse risk.

1111 **12. Licenses for existing assets**

1112 Question: Are the creators or original owners of assets (e.g., code, data, models), used in  
1113 the paper, properly credited and are the license and terms of use explicitly mentioned and  
1114 properly respected?

1115 Answer: [Yes]

1116 Justification: All evaluated models are cited with their original publications and checkpoint  
1117 sources. The dataset is released under CC BY 4.0; the toolkit under MIT. Overlap with YCB  
1118 and FewSQL object sets is acknowledged with citations.

1119 **13. New assets**

1120 Question: Are new assets introduced in the paper well documented and is the documentation  
1121 provided alongside the assets?

1122 Answer: [Yes]

1123 Justification: The dataset is accompanied by a Croissant metadata file (machine-readable  
1124 dataset card), a HuggingFace dataset card, and the rpx-benchmark toolkit with auto-  
1125 generated API documentation from source-level docstrings. The collection protocol, consent  
1126 details, privacy considerations, and known biases are documented in the dataset card.

1127 **14. Crowdsourcing and research with human subjects**

1128 Question: For crowdsourcing experiments and research with human subjects, does the paper

1129 include the full text of instructions given to participants and screenshots, if applicable, as

1130 well as details about compensation (if any)?

1131 Answer: [Yes]

1132 Justification: Data capture and annotation were performed by lab personnel (not crowd-

1133 sourced). The annotation protocol (SAM2 auto-masks with iterative human refinement) is

1134 described in Section 4. All participants provided informed consent for data collection and

1135 public release.

1136 **15. Institutional review board (IRB) approvals or equivalent for research with human**

1137 **subjects**

1138 Question: Does the paper describe potential risks incurred by study participants, whether

1139 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

1140 approvals (or an equivalent approval/review based on the requirements of your country or

1141 institution) were obtained?

1142 Answer: [N/A]

1143 Justification: The study involves only video capture of tabletop objects and human hands

1144 during manipulation. No experimental interventions on human subjects were conducted.

1145 Participants are lab members who consented to hand appearance in the interaction-phase

1146 recordings.

1147 **16. Declaration of LLM usage**

1148 Question: Does the paper describe the usage of LLMs if it is an important, original, or

1149 non-standard component of the core methods in this research? Note that if the LLM is used

1150 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

1151 scientific rigor, or originality of the research, declaration is not required.

1152 Answer: [N/A]

1153 Justification: LLMs are not part of the core method. LLMs were used only for writing

1154 assistance and code generation during toolkit development, which does not impact the

1155 scientific methodology or experimental results.